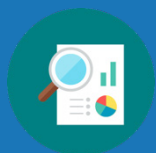


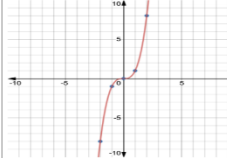
Unit 1 Overview: Introduction to Functions

Unit Narrative



In this unit, students will first be introduced to parent functions that are in Clarification 2 of MA.912.F.1.1. Students use multiple representations of these functions. The unit then continues with the concept of a function, working with rational exponents and radical expressions. Students will learn new and essential vocabulary such as parent functions, function notation, rational exponents, index, radicand, and radical expressions. Parent functions include linear functions, quadratic functions, cubic functions, square root function, cube root function, absolute value function, exponential function (growth), and exponential function (decay).

Section 1 introduces students to the parent functions, function notation, and interpreting function notation when given a context. Students become familiar with a variety of representations of parent functions. Section 1 will conclude with students extending their knowledge of functions from Grade 8 to evaluating functions in a mathematical and real-world context. Section 1 provides a broad introduction to functions and their representations in an effort to familiarize students with the types of functions they may encounter throughout the course. Section 1 also provides plenty of opportunities to establish class norms.

Cubic Function																
Parent Function	Table	Graph														
$y = x^3$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-8</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>8</td></tr><tr><td>3</td><td>27</td></tr></table>	x	y	-2	-8	-1	-1	0	0	1	1	2	8	3	27	
x	y															
-2	-8															
-1	-1															
0	0															
1	1															
2	8															
3	27															

In Section 2, students extend what they have learned about the Laws of Exponents in Grade 8 to numerical expressions that involve rational exponents. Students will work on evaluating these expressions and generating equivalent expressions. Students will learn the relationship between a rational exponent and a radical expression.

Section 3 extends the learning of Section 2 from rational exponents to radical expressions. This section begins with an overview and introduction into radical expressions by relating the Power of a Product Law to radicals to write an equivalent numerical radical. The section then transitions students into operations with radical expressions, including addition, subtraction, multiplication, and division of radical expressions.

Consider what differentiation may look like in this unit. Section 2 requires students to extend their learning from grade 8 with the Laws of Exponents. If students have taken Pre-Algebra as a prior course, they will have some background knowledge of the concept. If students have not taken Pre-Algebra prior to this course, students may have little to no background knowledge of this concept and additional support will be necessary.

Unit At-A-Glance

Section 1: Introduction to Functions	Benchmarks Addressed	Lesson 1	Classifying Parent Functions
	MA.912.F.1.1 MA.912.F.1.2	Lesson 2	Evaluating Functions in Function Notation
		Lesson 3	Interpreting Function Notation in the Real World
Section 2: Rational Exponents	Benchmark Addressed	Lesson 4	Applying the Laws of Exponents
	MA.912.NSO.1.1	Lesson 5	Exploring Rational Exponents
		Lesson 6	Writing Equivalent Expressions Using Rational Exponents
		Lesson 7	Evaluating Exponential Expressions
Section 3: Radical Expressions	Benchmark Addressed	Lesson 8	Equivalent Radical Expressions
	MA.912.NSO.1.4	Lesson 9	Adding and Subtracting Radical Expressions
		Lesson 10	Multiplying and Dividing Radical Expressions



Differentiation

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Engagement <i>The “WHY” of learning</i>	- For Career Exploration activities, consider doing a Virtual Field Trip with a person who works with that job or industry. This way, students can ask questions live and explore more about that specific career. - Consider turning the maze activity in Lesson 7 into a physical maze around the classroom for students to find their way out using the same problems from the maze in the book.
Representation <i>The “WHAT” of learning</i>	- Consider the use of anchor charts or visual reference guides when introducing the concept of parent functions. This way, students are able to refer back to them as they continue to work with them throughout the course. - For 1.10.2 Guided Instruction question 4, allow for additional physical movement for kinesthetic learners, consider designating one side of the room as “agree” and one side as “disagree.” Have students choose a side of the room in relation to Jean’s explanation, and have a representative from each group explain why they agree or disagree with Jean’s explanation. Repeat this process for Marcus’s explanation as well.
Action & Expression <i>The “HOW” of learning</i>	- For Wrap-Up activities that have students engage in self-reflection, consider the use of a technology tool such as FlipGrid to have students responds verbally and to one another. - For 1.10.3 Activity (Pass Around), consider having physical cards made for this activity for a hands-on experience for kinesthetic learners. These cards can be made by printing the page and cutting them out, or putting the problems directly onto index cards. Another alternative is to put the problems on chart paper and have students rotate to complete the next step of the given problem on the chart.

ELL Information

Suggested Instructional Strategies

- Use *Mark Up Text* to help students develop strategies for approaching more complicated word problems:
 - Teacher will use a word problem where students will be able to identify which parts would need to be represented in numbers, and how to write those numbers in an expression or equation.
 - Teacher will provide students with the following symbols.
 - ✓ - background knowledge, they know what to do
 - ? - confusing information; is this needed to solve the problem?
 - ! - marks where a variable is needed to represent a number
 - + - extra info in word problem; not needed
 - Using a small piece of text, teacher will model for students how you would Mark Up before releasing them to do the same.
 - Students will read the text slowly and carefully, focusing on one paragraph at a time and marking text based on the key. Then, they will turn to a partner to discuss each paragraph and compare where they made marks.
 - In closing, teacher will select students to share out their marks with the class at the end of the activity.

All glossary terms are available in multiple languages including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

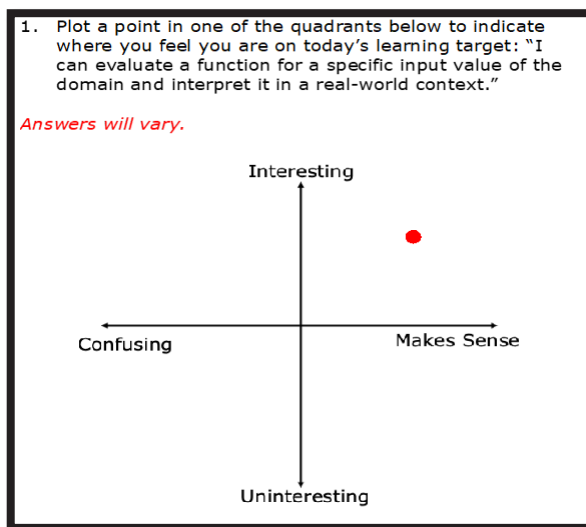
- Function
- Principal square root
- Relation
- Function notation

Additional Differentiated Support

Scaffolding Resources

Consider the learning styles and needs of your students when developing vocabulary and conceptual understanding from this section. Scaffolding in this section should include opportunities for students to apply reading and interpreting strategies that best fit the context of linear functions.

- Add an additional real-world problem that has a stronger cultural connection to your specific student population. For example, in 1.3.3 Guided Practice, the context is grizzly bears, which students may not be able to connect to with personal connection. Consider finding data of a context that is culturally appropriate for your student population.
- Provide Table cards on student desks for students to consistently refer to and apply when decoding and interpreting meaning from functions. This will be particularly helpful in Lesson 3, as students engage with real-world problems.
- Group or partner students intentionally considering their learning styles, particularly pairing visual and auditory learners. Students whose strengths lie in making use of pattern or structure visually pair well with students whose strengths lie in describing in meaningful context. One student might be able to identify domain and range from the graph, while the other student would then be able to successfully interpret and describe what that might mean in context.



Lesson 3 Wrap-Up includes qualitative feedback (pictured to the left) as a data point for teachers in this section. Consider incorporating both qualitative and quantitative feedback for each lesson to give students multiple entry points to provide data needed for remediation and scaffolding. Consider including specific learning targets from this section to better isolate students' gaps in learning.

On-Ramp

Use the videos and practice problems in the On-Ramp to Algebra 1 personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish.

- Identifying Rational Numbers
- Simplifying Square Roots
- Simplifying Cube Roots
- Writing Equivalent Radical Expressions
- Product Property
- Quotient Property
- Zero Exponent
- Exponent of One
- Negative Exponent
- Product to Power Property
- Power to Power Property



Section Overview

Section 1: Introduction to Functions (Lessons 1-3)

Benchmarks Addressed	Learning Targets	
MA.912.F.1.1 Given an equation or graph that defines a function, determine the function type. Given an input-output table, determine a function type that could represent it. <i>Clarification 1: Within the Algebra 1 course, functions represented as tables are limited to linear, quadratic, and exponential.</i> <i>Clarification 2: Within the Algebra 1 course, functions represented as equations or graphs are limited to vertical or horizontal translations, or reflections over the x-axis of the following parent functions: $f(x) = x$, $f(x) = x^2$, $f(x) = x^3$, $f(x) = \sqrt{x}$, $f(x) = \sqrt[3]{x}$, $f(x) = x$, $f(x) = 2^x$ and $f(x) = \left(\frac{1}{2}\right)^x$.</i> MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output. <i>Clarification 1: Problems include simple functions in two variables such as $f(x, y) = 3x - 2y$.</i> <i>Clarification 2: Within the Algebra 1 course, functions are limited to one-variable such as $f(x) = 3x$.</i>	Lesson 1	I can classify families of functions from a graph or equation.
	Lesson 2	I can evaluate a function for a specific input value of the domain.
	Lesson 3	I can evaluate a function for a specific input value of the domain and interpret it in a real-world context.

Section 2: Rational Exponents (Lessons 4-7)

Benchmark Addressed	Learning Targets	
MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents. <i>Clarification 1: Instruction includes the use of technology when appropriate.</i> <i>Clarification 2: Refer to the K-12 Formulas (Appendix E) for the Laws of Exponents.</i> <i>Clarification 3: Instruction includes converting between expressions involving rational exponent and expressions involving radicals.</i> <i>Clarification 4: Within the Mathematics for Data and Financial Literacy course, it is not the expectation to generate equivalent numerical expressions.</i>	Lesson 4	- I can apply the Laws of Exponents to write equivalent numerical expressions. - I can evaluate square roots and cube roots.
	Lesson 5	- I can apply the Laws of Exponents to rational exponents. - I can convert between expressions involving rational exponents and expressions involving radicals.
	Lesson 6	- I can apply the Laws of Exponents to rational exponents. - I can convert between expressions involving rational exponents and expressions involving radicals.
	Lesson 7	- I can evaluate numerical expressions involving rational exponents. - I can generate equivalent numerical expressions involving rational exponents.

Section 3: Radical Expressions (Lessons 8-10)

Benchmark Addressed	Learning Targets	
MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals. <i>Clarification 1: Within the Algebra 1 course, expressions are limited to a single arithmetic operation involving two square roots or two cube roots.</i>	Lesson 8	- I can relate the power of a product law of exponents to the product law of radicals - Given a numerical radical, I can use the product law of radicals to write an equivalent numerical radical.
	Lesson 9	I can add or subtract two square roots or two cube roots.
	Lesson 10	I can multiply and divide two square roots or two cube roots.



Vertical Alignment

8th Grade Unit 4
Equivalent Numerical
Expressions



Algebra 1 Unit 1
Introduction to
Functions



Algebra 2
Unit 1 and Unit 6

Prior Course(s)	Course Level	Future Course(s)
MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation. MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency. MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.	MA.912.F.1.1 Given an equation or graph that defines a function, determine the function type. Given an input-output table, determine a function type that could represent it. MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output. MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents. MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.	MA.912.F.1.1 Given an equation or graph that defines a function, determine the function type. Given an input-output table, determine a function type that could represent it. MA.912.NSO.1.5 Add, subtract, multiply, and divide algebraic expressions involving radicals.

1.1 Classifying Parent Functions

Lesson Introduction

 Benchmark	MA.912.F.1.1 Given an equation or graph that defines a function, classify this function type. Given an input-output table, determine a function type that could represent it.			 Learning Target	I can classify families of functions from a graph or equation.
 Benchmark Coherence	Building On MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.			Working Toward MA.912.F.1.1 This benchmark begins in Algebra 1 and extends toward function types beyond the clarifications included within this benchmark toward Algebra 2.	
 Essential Questions	- How can a function be transformed based on given values? - How can I use features of a graph to classify its function type?				
 Lesson Narrative	In this lesson, students are introduced to the concept of a parent function in each of the eight function types for the Algebra 1 course. Students will engage in a gallery walk to review the function, table, and graph representations of each type. They will be introduced to transformations of graphs using one of the parent functions in this lesson.				
 Component Summary	1.1.1 Warm-Up (~5 min.)	Would You Rather? Icebreaker (SE p. 3)	1.1.3 Guided Practice (10 min.)	Review given functions and determine how each one should be classified based on the function or graph, and learn about translations and reflections of graphs (SE p. 8)	
	1.1.2 Activity (20 min.)	Gallery walk that explores the eight different types of parent functions (SE p. 4)	1.1.4 Your Turn! (5 min.)	Determine the function family for a given equation or graph (SE p. 9)	
	1.1.5 Wrap-Up (~5 min.)	Self-reflection on their expertise with various math skills and growth mindset statements. (SE p. 9)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Translations of Functions		- Chart Paper - Markers (Optional) Sticky Notes		1.1.2 Activity: Gallery Walk – Prepare 8 posters that have the name of the parent function, the equation, a table of values, and its graph. Place posters around the room. Consider placing posters like Exponential Growth and Exponential Decay side-by-side.

 Teacher Tips	Common Misconceptions - Students may mistake the types of functions based on function features. - Students may mistake horizontal and vertical translations. - Students may mistake the exponential function types.	Things to Consider - Students may not be familiar with each other (or with you). Consider utilizing the Gallery Walk as an opportunity to communicate, establish, and practice classroom norms and discussion norms. Consider providing sentence stems to use during the Gallery Walk to build positive opportunities to engage. - This lesson could easily be divided up in different ways so that while establishing class norms for your class, students are also putting these norms into practice. - The purpose of this lesson is for the student to become familiar with the functions listed in clarification 2 of the benchmark MA.912.F.1.1. Students will continue to interact with most of these functions throughout the course. Full mastery of the parent functions is not an expectation of this lesson. - If time is an issue, then consider assigning 1.1.4 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Leverage 1.1.3 Guided Practice as a Notice and Wonder to explicitly activate the “noticing” that students expressed during the Gallery Walk. Students are prompted to notice what stayed the “same” to explicitly identify why a parent function remains in that family with translations.	MA.K12.MTR.2.1: Multiple Representations At some point during the lesson, prompt students to discuss the connection between the equation, the table of values, and the graph. Are these three different functions? Are they all the same function? Begin laying the foundation for multiple representations for students using this language.
	 Differentiate Instruction	Notice the varying levels of number sense and operations needed for different functions. The comprehension of each type of function can be an indicator of varying proficiency levels for your students to inform student grouping in upcoming lessons. Notice the varying entry points for 1.1.2 Activity. Students are associating locations with graphs and physically interacting, making sense of their “noticing” with a partner verbally, and purposefully making visual connections and using visual cues. Consider having students use that same number rubric in the Wrap-Up to label each of the 8 parent functions with their level of comprehension. Some students familiar with linear functions may struggle with cubic functions, indicating a gap in knowledge regarding exponents.
Lesson Notes:		

1.1.1 Warm-Up: Would You Rather?

Component Overview: Students will complete an introduction activity to get to know other students in the class by finding out what other students would rather do. This activity is designed to be done collaboratively by having students interact with a different student for each piece of the table.



1.1.1 Warm-Up: Would You Rather?

- Complete the chart by moving around the room and asking nine different people "Would you rather ...?" Introduce yourself to each person. Ask each person a different question from your chart. Record their name and their answer.

Answers will vary.

Would you rather ...

fly a kite or swing on a swing?	be a famous inventor or a famous writer?	go skiing or go to a water park?
be able to create a new holiday or create a new sport?	start a colony on another planet or be the leader of a small country on Earth?	own an old pirate ship with crew or a private jet with a pilot?
see a firework display or a circus performance?	be able to talk to land animals or talk to sea animals?	time-travel to the past or the future?



Prior to starting the activity, consider reinforcing any classroom management protocols that you have established as part of this activity. Be as explicit as possible with your instructions. Consider setting a time limit for this activity and reinforcing that time limit to maintain momentum for this lesson.



As an activity modification, consider creating two sides to the room: one side that represents choice 1 and one side that represents choice 2. Narrate the choices for the "would you rather" activity and have students stand on the side they choose.



After completing the activity, consider having students stay with their last partner. Have students introduce their last partner by sharing something the student would rather do. Repeat this for each student pair until all students have had a chance to be introduced to their peers.

1.1.2 Activity: Parent Functions Gallery Walk

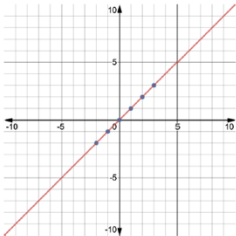
Component Overview: Print and paste (or write and sketch) each function on a poster with its equation, table of values, and graph on a poster with both prompts provided. Place the 8 posters around the room. Students can travel in pairs and choose a poster, or you can set a timer and have each pair spend a certain amount of time before rotating to a specific next poster. Consider labeling each poster with a letter (A-H), and using those letters as anchors. After students have completed all 8 rotations or visited all 8 posters, prompt them to revisit and draw any stars on their favorite “noticing” or “wondering.”



1.1.2 Activity: Parent Functions Gallery Walk

The study of functions is an area of focus in algebra. Functions are classified into families. The **parent function** is the simplest function of each family that preserves the characteristics of the family.

Here are eight parent functions that will be introduced in Algebra 1. After you examine each parent function, answer the questions that follow.

Linear Function																
Parent Function	Table	Graph														
$y = x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table>	x	y	-2	-2	-1	-1	0	0	1	1	2	2	3	3	
x	y															
-2	-2															
-1	-1															
0	0															
1	1															
2	2															
3	3															
What do you notice about the graph ?	<i>Answers will vary. The graph is a straight line. The graph begins in the bottom left and goes to the upper right.</i>															
What do you notice about the equation ?	<i>Answers will vary. The equation contains only one 'y' and one 'x'. There are no coefficients.</i>															



Potential Instructions (adjust as needed):

“With your partner, you will bring a post-it note and a pen. At your poster, make at least one noticing about the graph and at least one noticing about the equation. There are no right or wrong answers. After the first 3 rotations, you may notice everything you thought of has been said already. If this happens, instead place a check mark on your noticing, OR write something you wonder about the graph or a question you have.”



If you will be doing the activity with multiple classes, consider using post-it notes instead of having students write directly on the poster paper.

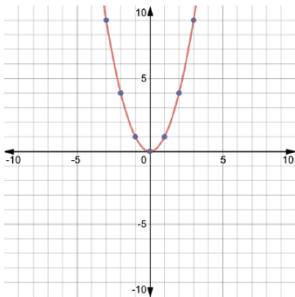


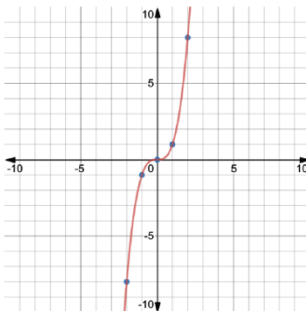
Consider asking prompting questions to guide students to identify unique features that will better position them for success in the following activity.

Example: Linear Function

- What do you notice about the relationship between the ordered pairs on the graph? (reference the table of values if needed)

1.1.2 Activity: Parent Functions Gallery Walk (continued)

Quadratic Function																
Parent Function	Table	Graph														
$y = x^2$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>9</td></tr></table>	x	y	-2	4	-1	1	0	0	1	1	2	4	3	9	
x	y															
-2	4															
-1	1															
0	0															
1	1															
2	4															
3	9															
What do you notice about the graph ?	<i>Answers will vary. The graph is not a straight line. The graph comes down and then shifts to go back up.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x value is raised to the second power. The equation is not linear.</i>															

Cubic Function																
Parent Function	Table	Graph														
$y = x^3$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-8</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>8</td></tr><tr><td>3</td><td>27</td></tr></table>	x	y	-2	-8	-1	-1	0	0	1	1	2	8	3	27	
x	y															
-2	-8															
-1	-1															
0	0															
1	1															
2	8															
3	27															



Consider what elements of a quadratic function students could identify as unique.

- What do you notice about where the graph “begins” or “ends” on the plane? How might that help you classify this as a quadratic function?
- What about the equation? What unique identifier tells you which type of function it is?
- Also, how does the exponent better inform why it “begins” and “ends” the way it does?

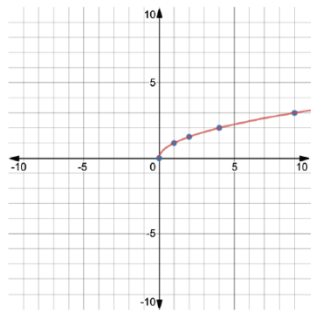


Consider what elements of a cubic function students could identify as unique.

- What do you notice about where the graph “begins” or “ends” on the plane? How might that help you classify this as a cubic function?
- What about the equation? What unique identifier tells you which type of function it is?

1.1.2 Activity: Parent Functions Gallery Walk (continued)

What do you notice about the graph ?	<i>Answers will vary. The graph starts going up and then 'flips' and continues going up. The graph isn't a line and has a 'swoosh' shape.</i>
What do you notice about the equation ?	<i>Answers will vary. The x value is raised to the third power. The equation is not linear.</i>

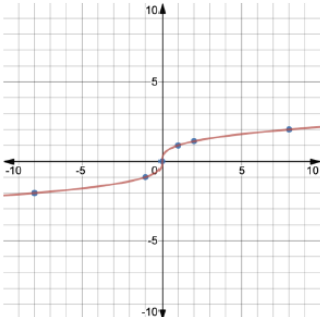
Square Root Function																
Parent Function	Table	Graph														
$y = \sqrt{x}$	<table><tr><th>x</th><th>y</th></tr><tr><td>-1</td><td>not real</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1.414</td></tr><tr><td>4</td><td>2</td></tr><tr><td>9</td><td>3</td></tr></table>	x	y	-1	not real	0	0	1	1	2	1.414	4	2	9	3	
x	y															
-1	not real															
0	0															
1	1															
2	1.414															
4	2															
9	3															
What do you notice about the graph ?	<i>Answers will vary. The graph starts at the origin and then goes up and to the right in a steep manner. The graph is not a straight line, but instead has a increasing curve.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x value is under the square root symbol. The equation is not linear.</i>															

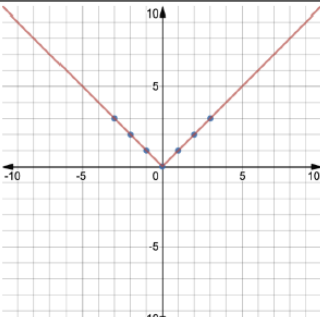


Consider what elements of a square root function students could identify as unique.

- Notice the square root in the equation. What impact does that have on the way the graph looks? Give an example.

1.1.2 Activity: Parent Functions Gallery Walk (continued)

Cube Root Function																
Parent Function	Table	Graph														
$y = \sqrt[3]{x}$	<table><tr><th>x</th><th>y</th></tr><tr><td>-8</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1.259</td></tr><tr><td>8</td><td>2</td></tr></table>	x	y	-8	-2	-1	-1	0	0	1	1	2	1.259	8	2	
x	y															
-8	-2															
-1	-1															
0	0															
1	1															
2	1.259															
8	2															
What do you notice about the graph ?	<i>Answers will vary. It begins on the left and shifts towards the origin to 'flip' and continue moving onward. I notice the left and right part of the graph look like reflections of each other.</i>															
What do you notice about the equation ?	<i>Answers will vary. I notice that the equation is not linear.</i>															

Absolute Value Function																
Parent Function	Table	Graph														
$y = x $	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>2</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table>	x	y	-2	2	-1	1	0	0	1	1	2	2	3	3	
x	y															
-2	2															
-1	1															
0	0															
1	1															
2	2															
3	3															

?

Consider what elements of a cube root function students could identify as unique, particularly to distinguish from a quadratic or square root function.

- Cover up one half of the coordinate plane, either the left or the right. What function does this look like? (square root)
- So, if you uncover that half of the plane, if half is a square root, what do you think the full function might be? (fourth?)
- Why is it a third, or cube, root function? (positive values are reflected in Quadrant 3)
- If you are only given the equation, how could you identify this from other types of functions?

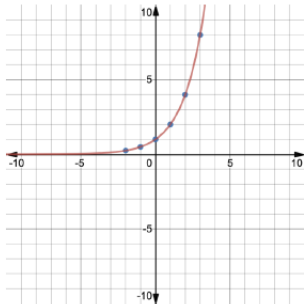
?

Consider what elements of an absolute value function students could identify as unique.

- Why do you think the graph makes an exact V-shape? Notice anything unique about the table of values that helps? (symmetry)
- What key feature in the equation can serve as a unique identifier?

1.1.2 Activity: Parent Functions Gallery Walk (continued)

What do you notice about the graph ?	<i>Answers may vary. The graph makes a v shape. The graph comes down from the left and then goes up and to the right once it hits the origin.</i>
What do you notice about the equation ?	<i>Answers may vary. The x value is within absolute value bars. This equation is not linear.</i>

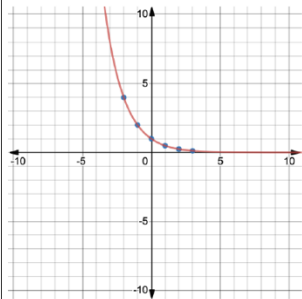
Exponential Function																
Parent Function	Table	Graph														
Exponential Growth Base (2) $y = 2^x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>$\frac{1}{4}$</td></tr><tr><td>-1</td><td>$\frac{1}{2}$</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>8</td></tr></table>	x	y	-2	$\frac{1}{4}$	-1	$\frac{1}{2}$	0	1	1	2	2	4	3	8	
x	y															
-2	$\frac{1}{4}$															
-1	$\frac{1}{2}$															
0	1															
1	2															
2	4															
3	8															
What do you notice about the graph ?	<i>Answers will vary. The graph begins with a narrow increase and then increases sharply. The graph goes up slowly and then shoots up quickly.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x-value is in the exponent place. This equation is not linear.</i>															



Consider what elements of an exponential growth function students could identify as unique, particularly the difference between growth and decay.

- Compare this graph to exponential decay. Where does this graph “start” and where does it “end”?
- What does your response tell you about why this function represents growth? (use the table of values to scaffold this question)
- What about the equation? What is the same or different compared to decay?

1.1.2 Activity: Parent Functions Gallery Walk (continued)

Exponential Function																
Parent Function	Table	Graph														
Exponential Decay Base $\left(\frac{1}{2}\right)$ $y = \left(\frac{1}{2}\right)^x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>2</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>$\frac{1}{2}$</td></tr><tr><td>2</td><td>$\frac{1}{4}$</td></tr><tr><td>3</td><td>$\frac{1}{8}$</td></tr></table>	x	y	-2	4	-1	2	0	1	1	$\frac{1}{2}$	2	$\frac{1}{4}$	3	$\frac{1}{8}$	
x	y															
-2	4															
-1	2															
0	1															
1	$\frac{1}{2}$															
2	$\frac{1}{4}$															
3	$\frac{1}{8}$															
What do you notice about the graph ?	<i>Answers will vary. The graph starts from the left and decreases quickly. This one looks like the opposite of the exponential growth.</i>															
What do you notice about the equation ?	<i>Answers will vary. The x-value is in the exponent and a fraction is the base. The equation is not linear.</i>															



Consider what elements of an exponential decay function students could identify as unique, particularly the difference between growth and decay.

- Compare this graph to exponential growth. Where does this graph “start” and where does it “end”?
- What does your response tell you about why this function represents decay? (use the table of values to scaffold this question)
- What about the equation? What is the same or different compared to growth?

1.1.3 Guided Practice: Classifying Functions

Component Overview: Students analyze the equations and corresponding graphs of functions to determine the impact of translations of classification. Teachers should monitor student responses in questions 1 and 2. Students apply their understanding by classifying more functions in question 3.



1.1.3 Guided Practice: Classifying Functions

Functions can be transformed by translations, dilations, and reflections.

In this example, the quadratic parent function is translated vertically, horizontally, and reflected over the x -axis.

Quadratic Parent Function	Translated Vertically	Translated Horizontally	Reflected over the x -axis
$y = x^2$	$y = x^2 - 2$	$y = (x - 5)^2$	$y = -(x)^2$

- What stayed the same in each of the equations?
In each equation, the y -value remained on the left hand side of the equation. In each equation, the exponent remained 2.
- What stayed the same in each of the graphs?
In each graph, the size of the parabola (width and height) remained the same.



Notice and Wonder

Utilize a Notice and Wonder routine to position students to formalize specific learning targets. Frame questions 1 – 2 by prompting students to specifically notice how the graph changed in relation to how the equation changed.

- The equation of each function retains an exponent of 2. Students should formalize this exponent as the key feature that can classify a quadratic.
- The “shape” of the graph stays the same. Students should formalize that translated functions still belong to the same family of functions.

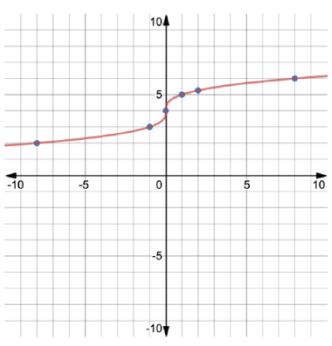
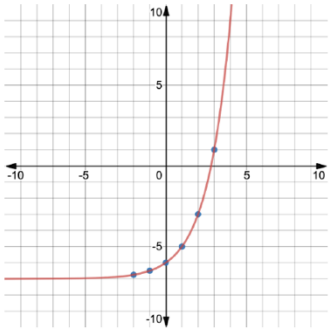


Consider prompting questions to target specific learning outcomes for these questions.

- Is this function still a quadratic function? (For example, “This is still a quadratic function just with a different vertex.”)
- Can you restate what you mean by “move up” but use the word “translate” instead? (“The second function is still a quadratic, but the -2 just means the function was translated 2 units down.”)

1.1.3 Guided Practice: Classifying Functions (continued)

3. Classify each function into its function family based on the characteristics observed for the parent functions.

Function	Function Family
$y = x + 9 $	<i>Absolute Value</i>
$y = -\left(\frac{1}{2}\right)^x$	<i>Exponential Decay</i>
$y = x^3 - 6$	<i>Cubic</i>
	<i>Cube Root</i>
	<i>Exponential Growth</i>



Consider incorporating the Gallery Walk posters as scaffolding for students. Prompt students to answer each question first, and then provide a reason why they chose that function. Students should then visit the poster of their answer to verify their answer.



Consider modeling think-aloud questions to provide students with questions to consider.

- What makes this equation unique?
- What does that feature tell me about the type of function it belongs to?
- Does that have anything to do with what it looks like on the graph?

1.1.4 Your Turn!

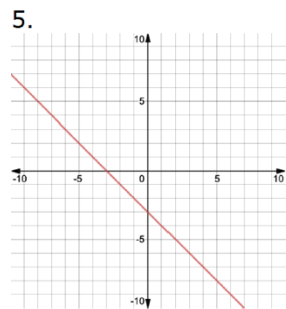
Component Overview: Students will independently practice determining the function family given an equation or graph.



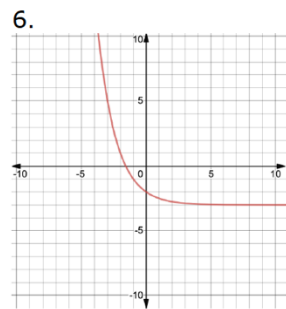
1.1.4 Your Turn!

Classify the function family for each equation or graph.

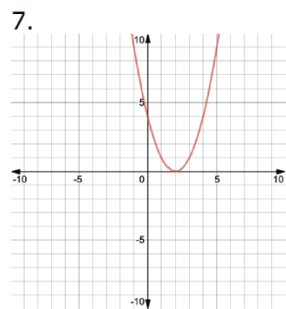
1. $y = (x + 2)^3$ Function Family: Cubic
2. $y = -x - 23$ Function Family: Linear
3. $y = 2^{(x-7)}$ Function Family: Exponential Growth
4. $y = (x)^2 + 31$ Function Family: Quadratic



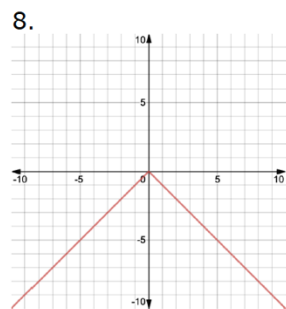
Function Family:
Linear



Function Family:
Exponential Decay



Function Family:
Quadratic



Function Family:
Absolute Value



Students have not formally learned about most of the functions presented. Students might struggle most with identifying the unique differences between functions that look similar on a graph (i.e., answering “square root function” to question 6).



Consider prompting questions that identify the unique differences between these functions.

- Question 1: What is the value of the exponent? What does that tell you about the function?
- Question 3: Where is the variable x ? What does that tell you about the type of function?
- Question 6: Note the square root function. What is different about this function?

1.1.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection where they rate themselves on their expertise with various math skills and growth mindset statements. This activity is designed to be done independently to get an accurate self-reflection from each student.



1.1.5 Wrap-Up: Reflection

- As you begin your journey through Algebra 1, rate yourself on where you feel you are today on the following skills:

Answers will vary.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can evaluate expressions using order of operations.	
I can graph points on a coordinate plane.	
I can apply the laws of exponents.	
I can determine whether a relation is a function.	

- Rate yourself based on how you feel about the following statements:

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am willing to try new things.	
I persevere when something is challenging.	
I am a student learner.	
I am a mathematician.	



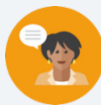
Notice these statements are not related to this lesson but meant to collect a topical pulse check on a few concepts students will encounter in the course.

Alternatively, consider having students number each activity in this lesson with the same rubric, or number each type of parent function in this lesson with their level of understanding.

1.2 Evaluating Functions in Function Notation

Lesson Introduction

 Benchmark	MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.		 Learning Target	I can evaluate a function for a specific input value of the domain.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.		MA.912.F.3.3 Solve mathematical and real-world problems involving functions that have been combined using arithmetic operations.	
 Essential Questions	- How can I determine if a relation is a function? - What is the difference between evaluating $f(5)$ and $f(x) = 5$?			
 Lesson Narrative	In this lesson, students are introduced to the concept of a function. They will learn the key vocabulary associated with functions. Additionally, students will apply this knowledge to evaluate functions provided in the form of algebraic expression, graph, or mapping diagram.			
 Component Summary	1.2.1 Warm-Up (~5 min.)	Food science career exploration activity (<i>SE p. 10</i>)	1.2.4 Guided Instruction (15 min.)	Evaluating functions activity (<i>SE p. 12</i>)
	1.2.2 Collaboration (5 min.)	Introduction to the concept of a function, represented in multiple ways (<i>SE p. 11</i>)	1.2.5 Try It (5 min.)	Evaluating functions provided in multiple forms practice (<i>SE p. 14</i>)
	1.2.3 Exploration (5 min.)	Exploration of the different graphs of functions, noticing basic behaviors of each one (<i>SE p. 11</i>)	1.2.6 Your Turn! (5 min.)	Evaluating functions provided in multiple forms independent practice (<i>SE p. 15</i>)
	1.2.7 Wrap-Up (~5 min.)	Ticket Out the Door on evaluating functions. (<i>SE p. 15</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Relation - Function <u>Additional Terms:</u> - Function Notation - Input - Output		(Optional) Chart Paper for Anchor Chart	Warm-Up 1.2.1 contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions - Students may not accurately differentiate between evaluating $f(5)$ vs $f(x) = 5$. - Students may make calculation errors in the evaluation. - Students may switch the input and output values on a graph. - Students may think repeating output values makes a relation not a function.	Things to Consider This lesson contains a lot of new vocabulary and concepts. Consider creating a foldable or guided notes document for students to use as a later reference with examples of different representations.
	Literacy and Language Routines Compare and Connect 1.2.4 Guided Instruction provides students with an opportunity to receive guided instruction on how to evaluate functions, represented in several forms. Consider leveraging the Compare and Connect protocol during this activity. For each question set, model one question by thinking out loud and questioning why it might be done this way, prompting students to reflect and respond prior to being released to try on their own.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency 1.2.5 Try It has students working to practice evaluating functions in multiple forms on their own. Upon completion, they will have their work checked by their partner. Each partner will provide the other with feedback about their final answer based on the input values and calculations. Students will use this feedback to improve their efficiency when performing calculations.
	1.2.5 Try It provides students with an opportunity to practice independently on evaluating functions. Student performance in this activity can drive potential differentiation in 1.2.6 Your Turn! if students may need additional support. Consider the use of an anchor chart or alternative when introducing this concept. Students may not be familiar with function language. Creating a visual aid for students to have can assist students with knowing what to do in a situation where $f(5)$ is being presented vs where $f(x) = 5$ is being presented. It can also assist students with knowing how to read and interpret functions. The 1.2.7 Wrap-Up for this lesson is a Ticket Out the Door that will provide formative feedback on how well any given student is performing towards mastery of the concepts in this lesson.	
 Differentiate Instruction		
Lesson Notes:		

1.2.1 Warm-Up: Food and Research Career

Component Overview: Play the Career Profile video on a Food and Research Career for the entire class. After students watch the video (1:22 minutes), prompt them to reflect on the profession by answering the two questions independently.



1.2.1 Warm-Up: Food and Research Career

Richard works in Research and Development for Kellogg's. One of the projects he is most proud of involves developing a line of organics products and no added sugar products. Richard also points out that working with a team provides an exciting variety of experiences that is crucial and rewarding to the work of the company.

1. How could mathematics help in the food science industry?
Answers will vary.
2. Why is it important to have teamwork skills when working for a company?
Answers will vary.



Have you ever heard of the food science industry? What would it be like if the person doing this job did not have teamwork skills?

1.2.2 Collaboration: Functions

Component Overview: Students will be introduced to the concept of a function, represented in multiple ways. For this activity, begin by reviewing the formal definitions of a relation and a function and review the three examples that will be used for this activity.



1.2.2 Collaboration: Functions

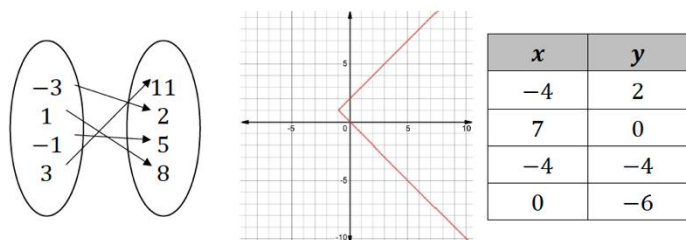


A **relation** is a set of input-output pairs.



A **function** is a mathematical relation for which each element in the domain corresponds to exactly one element in the range.

Three different relations are shown, with each represented in a different way.



- With your partner, write an explanation identifying which of the relations above are functions.
The first relation is a function. Each input has exactly one output.
The second relation is not a function. The x-values (input) have two different y-values (outputs).
The third relation is not a function. The x-values (input) have two different y-values (outputs).



Understanding the difference between a relation and a function is taught in grade 8, MA.8.F.1.1. This Collaboration is designed to help students recall the definition of a relation and a function. The benchmark, MA.912.F.1.2, does not require a student to distinguish between a relation and a function. Reinforcing the vocabulary can be done throughout the course.

Consider giving students an example of a relation that is not a function, and then transition to an example where the relation is a function.



- How did you use the definitions to determine if each relation was a function?
- Were there any representations that were easier to work with?

1.2.3 Exploration: Graphs of Functions

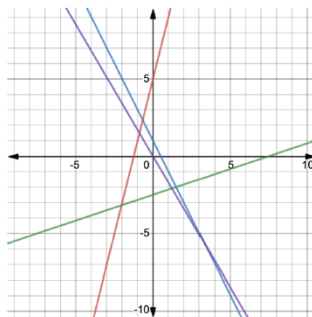
Component Overview: Students will explore the different graphs of functions, noticing basic behaviors of each one. For this activity, begin by introducing students to the context of the question, including matching each line with its corresponding equation using the matching colors. For questions 1 – 4, release students to explore these questions and answer them with a partner after the explanation of the context. Bring the class back together to review their answers to each question.



1.2.3 Exploration: Graphs of Functions

The equations and graphs of four linear functions are shown.

$$\begin{aligned} y &= 4x + 5 \\ y &= -2x + 1 \\ y &= \frac{1}{3}x - 2.5 \\ y &= -1.7x \end{aligned}$$



- Write the equations of the lines that decrease from left to right.
 $y = -2x + 1$
 $y = -1.7x$
- Write the equations of the lines that increase from left to right.
 $y = 4x + 5$
 $y = \frac{1}{3}x - 2.5$
- What do the equations of the lines that increase from left to right have in common?
Answers will vary. The slopes of the lines are positive. The coefficients of x are positive.
- How are the equations of the lines that decrease from left to right different from the equations of the lines that increase from left to right?
Answers will vary. The slopes of the lines are negative. The coefficients of x are negative.



The graph of each line and its corresponding equation are matching by using the same color. Draw attention to how students can identify the line that matches each equation. If students have difficulty seeing the colors, consider labeling each equation and then labeling their corresponding graph on the coordinate grid.



- Why do you think that all these questions use the phrase from “left to right?”
- Why do you think that it is important to have a standard way of reading a graph?



Allow students to share out their answers for each question. Highlight any misconceptions that are heard such as the way the graph was read. For questions 3 and 4, students may not use the vocabulary of coefficients or slope when indicating negative or positive. Clarify which value is negative or positive in both.

1.2.4 Guided Instruction: Evaluating Functions

Component Overview: Students will receive guided instruction on how to evaluate functions. For this activity, begin by providing students with an introduction into the vocabulary of a function including function notation, name of function, input of function, and output of function. Use the narrative information prior to question 1 to assist with providing this instruction.

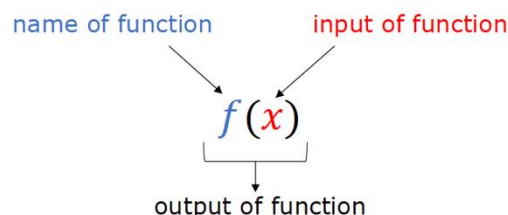


1.2.4 Guided Instruction: Evaluating Functions

Each of the functions in the Exploration starts with “ $y =$.” If asked to use the function y from the list of functions to find a specific value, it would not be clear which of the four functions to use.

Since mathematicians often work with multiple functions at a time, **function notation** is used to name specific functions.

The notation $y = f(x)$ defines a function named f . This indicates “ f is a function of x .” The letter x represents the input value, or independent variable. The letter y , or $f(x)$, represents the output value, or dependent variable. The notation $f(x)$ is read, “ f of x .”



A function can be named using any letter. For example, the previous functions can be renamed as shown:

$$f(x) = 4x + 5 \qquad g(x) = -2x + 1$$

$$h(x) = \frac{1}{3}x - 2.5 \qquad c(x) = -1.7x$$

Now, if asked to use the function h to find a specified value of x , it would be clear which function to use.



Consider the use of an anchor chart or alternative when introducing this concept. Students may not be familiar with function language. Creating a visual aid for students can assist them when presented with a situation such as $f(5)$ vs. $f(x) = 5$. It can also assist students in reading and interpreting the functions.



For the functions provided, model the first one for students on how to read it properly. Have students take turns reading the additional three to practice how to read each function correctly. Use these functions to reinforce the definition of a function that was introduced earlier in the lesson.

1.2.4 Guided Instruction: Evaluating Functions (continued)

Functions are evaluated in the same way equations are evaluated when given a value for a variable.
Consider the function $f(w) = -7w^2 - 10w + 3$.

- Find the value $f(10)$.

$$f(w) = -7w^2 - 10w + 3$$

$$f(10) = -7(10)^2 - 10(10) + 3$$

$$f(10) = -797$$

When the input is 10, the output of the function is -797 .

Using function notation, this is written $f(10) = -797$.

- Find the value $f(-3)$.

$$f(-3) = -30$$

This same function $f(w)$ can be represented in other ways.

Shown here are a table of values, a graph, and a mapping diagram. The function can be evaluated from each of these representations.

Table of Values

w	$f(w)$
-15	-1422
-10	-597
-1	6
2	-45

- Find $f(-10)$.

$$f(-10) = -597$$

- If $f(w) = -45$, what is the value of w ?
 $w = 2$



Prior to modeling question 1, review the function that is provided for the context for questions 1 and 2. Remind students of how this function is read, and what it means for an input value. Take a moment to remind students to take their time when evaluating to avoid making a calculation error with exponents or negative signs.

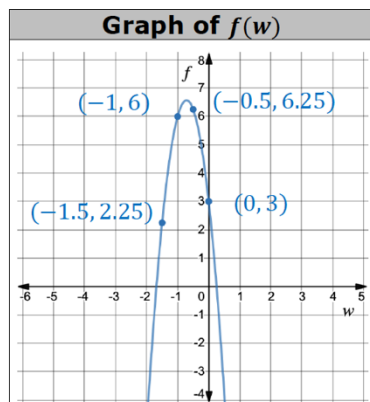


- Were there any areas that caused a calculation error when evaluating?
- Does the value you input change what the function does?



How is the process of evaluating the function different when evaluating for $f(-10)$ versus $f(w) = -45$?

1.2.4 Guided Instruction: Evaluating Functions (continued)



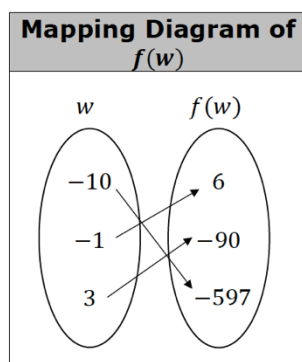
Find each value from the graph.

5. $f(-1.5) = 2.25$

6. $f(0) = 3$



Prior to modeling question 5, review the graph of the function using decreasing and increasing language that students used in the prior activity. This will help reinforce graphing protocols and language. When modelling question 5, connect the input value to the x -value and output value to the y -value.



Find each value from the mapping diagram.

7. $f(3) = -90$

8. Find the value of w when $f(w) = 6$.
 -1



Consider using a foldable to show students how one function can be represented in multiple ways. The top of the foldable can be the overall function and there would be three squares under it, each representing that function being represented in a table, graph, and mapping diagram form.

1.2.5 Try It: Function Notation

Component Overview: Students will practice evaluating functions provided in multiple forms. For this activity, provide students with instructions to complete each question on their own and check their work with a partner. After completing the activity consider bringing all students back together to review the answers and steps for each question.



1.2.5 Try It: Function Notation

1. Given the function $c(y) = -6y - 5$, find each value.

a. $c(-4) =$
19

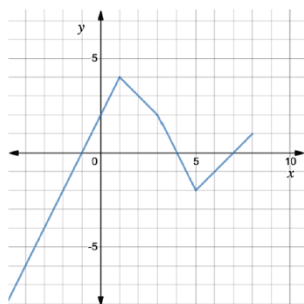
b. $c\left(\frac{3}{4}\right) =$
-9.5

c. For what value of y is $c(y) = 9$?
-2.3

2. Given $g(k) = 2k^2 - 4$, find $g(-5)$.
46

3. Given $h(x) = \sqrt{2.5x + 1}$, find $h(3.2)$.
3

4. Use the graph of $y = f(x)$ below to find each value.



a. $f(1) =$ **4**

b. $f(-2) =$ **-2**

c. For what value of x is $f(x) = -4$? **-3**



Student performance on these questions can provide data for what students are retaining from this lesson, and what additional supports may be necessary from prior learning deficits. Possible scenarios:

- Unable to input the correct value into the correct location in the function
- Calculation errors
- Unable to identify a solution using a graph



Watch for student answers to question 1 part c and question 4 part c. Many students who do well with evaluating functions mistake these types of questions where it isn't an input and evaluate scenario. If students are missing these two but are getting the others correctly, consider reviewing this concept with students again.



Allow students to share out their answer for each question. They should share their work to illustrate how they arrived at the final answer.

1.2.6 Your Turn!

Component Overview: Students will work independently to practice evaluating functions given different forms. For this activity, students should be released to complete questions 1 – 3 on their own. Call students back together to review the answers for each question.



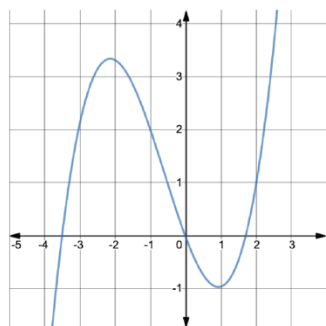
1.2.6 Your Turn!

1. Find each value given the function $m(t) = \frac{t}{3} + 6$.

a. $m(-9) =$
3

- b. Find the input for which $m(t) = 13$.
21

2. Find each value using the graph of the function $f(x)$.



a. $f(-3.5) =$ **0**

b. $f(2) =$ **1**

c. Find x when $f(x) = 4$.
2.5

3. Use the table of values representing $D(r)$ to find each value.

a. $D(-4) =$ **9**

b. $D(-6) =$ **9**

r	$D(r)$
-6	9
0	-5
-4	9
2	0
9	7

- c. Is $D(r)$ a function? Explain why or why not.
Yes, $D(r)$ is a function. For each input there is one unique output.



Consider using the time for this activity to provide differentiated support to students who may be struggling with the lesson content.

Use the information related to student performance from the prior activity, such as where they input the value and what they calculate, to determine which students may need to be pulled together for small group review.

1.2.7 Wrap-Up: Ticket Out the Door

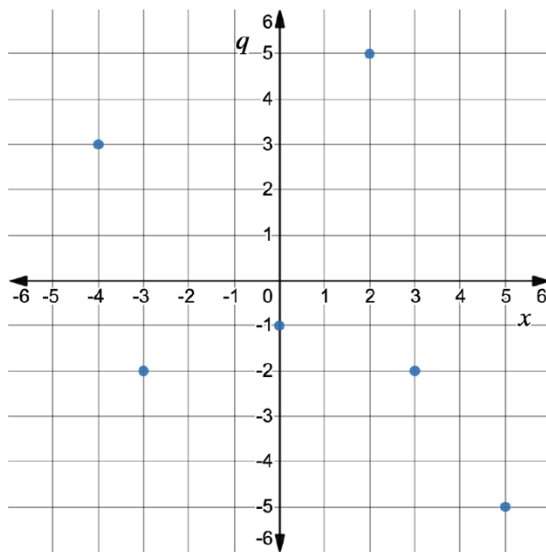
Component Overview: Students will complete a Ticket Out the Door that assesses their ability to evaluate functions. Each student will independently do this activity. The teachers will be able to get accurate data about the student's understanding of the lesson content.



1.2.7 Wrap-Up: Ticket Out the Door

- Given the function $h(t) = t^3 - 2t$, find $h(-7)$.
 $h(-7) = -329$

- Use the graph of $q(x)$ to find $q(-3)$.



$q(-3) = -2$

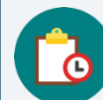


Function notation is used throughout the course. Use the information from this Wrap-Up to know how well your students understand evaluating functions algebraically and from a graph. As students continue to evaluate using functions throughout the course, it is more important that you look for the student to progress than to have full mastery at the end of this lesson.

For students who do not perform well on this Wrap-Up activity, determine if there is a part of the concept they may not understand. For example, students who get question 1 incorrect but question 2 correct may understand the concept of input values but may have calculation issues. Then, use this information to inform your teaching.

1.3 Interpreting Function Notation in the Real World

Lesson Introduction

 Benchmark	MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.		 Learning Target	I can evaluate a function for a specific input value of the domain and interpret it in a real-world context.
 Benchmark Coherence	Building On MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or a mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.		Working Toward MA.912.F.3.3 Solve mathematical and real-world problems involving functions that have been combined using arithmetic operations.	
 Essential Questions	- How can evaluating a function be mathematically accurate but not viable for a context? - What impact does the context of a function have on the reasonableness of the answer?			
 Lesson Narrative	In this lesson, students will continue with evaluating functions, adding in a new layer of interpretation. Students will work with functions that represent real-world scenarios to interpret values and assess their reasonableness based on the context.			
 Component Summary	1.3.1 Warm-Up (~5 min.)	Fill-in-the-blank activity to construct two functions that are both true (<i>SE p. 16</i>)	1.3.4 Try It (5 min.)	Evaluate and interpret a function that represents a real-world context (<i>SE p. 21</i>)
	1.3.2 Guided Instruction (15 min.)	Interpreting functions presented in multiple forms (<i>SE p. 17</i>)	1.3.5 Your Turn! (5 min.)	Evaluating and interpreting functions that represent a real-world scenario independently (<i>SE p. 21</i>)
	1.3.3 Guided Practice (15 min.)	Interpreting real world functions (<i>SE p. 19</i>)	1.3.6 Wrap-Up (~5 min.)	Self-assessment of interpreting functions (<i>SE p. 22</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Function - Input - Dependent		N/A	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider		
	- Students may not fully understand context to interpret it appropriately. - Students may evaluate function and forget to interpret if the value is reasonable. - Students may make calculation errors when evaluating functions.	- This lesson requires a lot of reading as each problem involves a real-world context. Consider leveraging word problem strategies (see Unit Overview for an example) that students can use to better understand the problems. - Interpreting real-world contexts is an expectation of many of the benchmarks in Algebra 1. Students may struggle with an interpretation because their experience with mathematics is to arrive at an answer but not to relate that to the context provided. Allow for students to have productive struggles while working through the contexts. Avoid establishing a strategy that may ultimately fail a student as the contexts and their mathematical representations change. Encourage students to share their thinking so that other students can consider different ways of looking at the contexts. - If time is an issue, then consider assigning 1.3.5 Your Turn! for homework.		
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Three Reads Each activity in the lesson (excluding 1.3.1 Warm-Up and 1.3.6 Wrap-Up) involve word problems that impact the work students will complete. Consider leveraging the “three reads” protocol. On the first read, students have a goal of understanding the context. On the second read, students have a goal of understanding the mathematics required. On the third read, the students generate a plan for solving the problem.		MA.K12.MTR.6.1: Reasonableness The core activities of this lesson allow students to learn and practice evaluating functions and interpreting values within the context provided. They will encounter values that are mathematically appropriate but are not reasonable for the context provided. Students will engage in assessing whether values are reasonable throughout the lesson.	
 Differentiate Instruction	1.3.4 Try It provides students with a structured practice opportunity. Student performance during this activity can determine whether they may need additional support during 1.3.5 Your Turn! when they practice individually. In this lesson, students will be asked to fill in the blank to describe the x – and y – axis. Consider having a blank graph available and having students label the graph to visually connect the description to the graph. 1.3.5 Your Turn! provides a valuable formative assessment opportunity to determine what students know about the content of this lesson. Students who perform poorly on this activity would need additional support for this content.			
Lesson Notes:				

1.3.1 Warm-Up: Equivalent Function Values

Component Overview: Students will complete a fill-in-the-blank activity to construct two functions that are both true, with digit use limitations. This activity is designed to be done independently.



1.3.1 Warm-Up: Equivalent Function Values

- Fill in the boxes to construct two functions where the final equality statement is true. Use the digits 1 to 9 once each at most.

$$g(x) = \boxed{}x - \boxed{}$$

$$g(\boxed{}) = 26$$

Answers will vary. Examples include:

$$g(x) = 9x - 1; g(3)$$

$$g(x) = 7x - 2; g(4)$$



Students may struggle with the task, as it will be their first time with this type of activity in the course. Consider spending extra time introducing what the expectations for this type of problem are, with particular emphasis on the digit limitations, as there will be many more of these problem types throughout the units.



Consider showing an example of filling in a box and eliminating that number from being used again. For visual learners, write each digit from 0 to 9 on a common location such as a whiteboard. When a digit is used, cross that number out so that visual learners can connect to the idea of that number no longer being available.



Allow students to share out their answers and determine who has correct answers. Have students share out the method(s) and/or strategies they used to arrive at their answer.

1.3.2 Guided Instruction: Interpreting Functions

Component Overview: Students will receive guided instruction on how to interpret a function presented in multiple forms and determine if the values found are reasonable within the context.



1.3.2 Guided Instruction: Interpreting Functions

Recall that functions have a special notation which allows unique relationships to be expressed in a concise and meaningful way.

The value of a function is dependent on the input. The statement $D(2) = 60$ means that for an input of 2, the value of the function, D , is 60.

Functions can be used to model real-world relationships.

1. If $F(t)$ models the outside temperature in degrees Fahrenheit t hours after 6 a.m., the statement $F(2) = 60$ has a specific meaning within the context.

$F(2) = 60$ can be interpreted as 2 hours after 6 a.m., which is 8 a.m., the outside temperature is 60 degrees Fahrenheit.

2. Physics students drop a ball from the top of a 400 foot high building and model its height above the ground as a function of time using the function $H(t) = 400 - 16t^2$. The height, H , is measured in feet and time, t , is measured in seconds.
 - a. Find the value of $H(0)$.
400
 - b. What does $H(0)$ mean in the context of this problem?
The height of the ball before any time has elapsed, or height of the building.
 - c. Evaluate $H(2)$.
336



To review what was learned in 1.2.4 Guided Instruction (Lesson 2), consider the following questions:

- What does the special notation for a function look like? What are some examples?
- What does it mean when $f(x)$, $g(x)$, $h(x)$, etc.?



Consider using question 1 as a modeling opportunity for thinking through a problem and determining its interpretation. Ask students while working through the problem about whether the interpretation is viable and appropriate, using evidence from the problem. Discuss how the interpretation would differ if the values in the function were reversed.



Student performance with question 2 can provide data for potential differentiation or support. Students who can do question 2 parts a and c, but struggle with b and d understand how to evaluate functions from prior lesson, but don't understand context and interpretation from current lesson.

1.3.2 Guided Instruction: Interpreting Functions (continued)

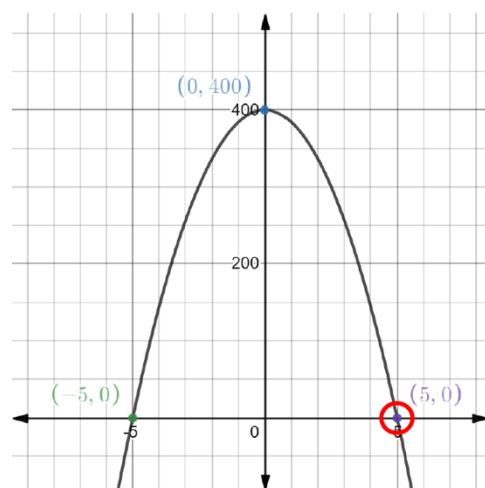
- d. Interpret the meaning of $H(2)$ based on the given context.

After 2 seconds has passed, the ball is 336 feet above ground.

- e. José is asked to interpret the value of $H(-3)$. Is interpreting this value reasonable in the context of this situation?

In this context, a negative value is not reasonable. Since t represents time in seconds, it cannot be negative.

3. The graph of $H(t)$ is shown on the coordinate grid.



- a. Based on the context, the x -axis represents *Time elapsed since dropping the ball* and the y -axis represents *the height of the ball from the ground*.

- b. What will the height of the ball be when it hits the ground?

0 feet

- c. Circle the point where the ball will hit the ground.

Answer on graph



For question 2 part e, consider engaging students in a conversation about reasonableness in this context. This can help reinforce the idea of interpreting the meaning, addressing potential misconceptions, and reinforcing the concept of viability. Ask students if there are any potential situations where the negative value would be reasonable. Be careful about establishing with students that “there is no such thing as negative time”. A negative value that represents time is establishing what may have occurred before. For some contexts this makes sense. For example, a student may consider that $H(-3)$ is when the ball was, three seconds before it was taken to the top of the building. This is a valid interpretation. If a student does not provide this interpretation, then consider including it for students to discuss.



For visual learners, consider taking the information from the blanks in part a of question 3, and having students label the graph as a group.



- Why is the circle only on the x -intercept at $(5,0)$?
- Why would the other x -intercept at $(-5,0)$ not be circled?
- What about point $(0,400)$? Why is it not circled?
- What does it mean in this context?

1.3.2 Guided Instruction: Interpreting Functions (continued)

- d. Write the point where the ball hits the ground using function notation.

$$H(5) = 0$$

- e. How long did it take the ball to hit the ground after it was released?

5 seconds

- f. Discuss with your partner what $H(1.8) > H(3.4)$ means in terms of the context.

The height of the ball at 1.8 seconds is greater than the height of the ball at 3.4 seconds. This is because the height of the ball, from the ground, decreases as the seconds increase because it is falling further and further from the top of the building.



- How can you determine how long it took the ball to hit the ground after it was released?
- How does the solution relate to the information that is present on the graph?



For question 3 part f, consider allowing students to share out what they believe this means. There are possible misconceptions with this question that could be addressed through a facilitated discussion.



For a challenge, have students reverse the inequality symbol and answer the same questions. Does the resulting interpretation make sense? Is it viable?

1.3.3 Guided Practice: Interpreting Functions

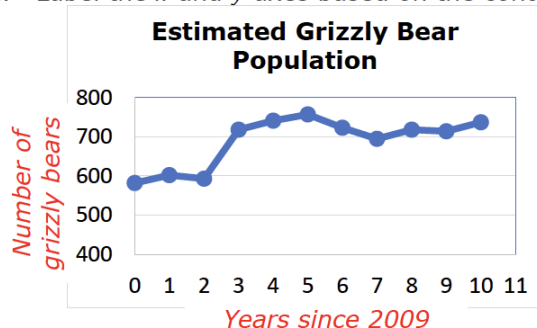
Component Overview: Students will receive a guided practice opportunity to evaluate and interpret real world functions.



1.3.3 Guided Practice: Interpreting Functions

1. The population of grizzly bears at Yellowstone National Park has been threatened with local extinction since 1975. Through many conservation efforts, grizzlies have made a remarkable recovery. The function $y = G(t)$ is modeled below, where t is the number of years since 2009 and G is the number of grizzly bears.

- a. Label the x and y axes based on the context.



- b. Find the approximate value from the graph above.

$$G(3) \approx 718$$

- c. In the context of this problem, $G(3)$ means that 3 years after 2009, there were approximately 718 bears in Yellowstone.

- d. Find the approximate value from the graph.

$$G(8\frac{1}{2}) \approx 710$$

- e. What does $G(8\frac{1}{2})$ mean in the context of this problem?

At $8\frac{1}{2}$ years after 2009 there would be approximately 710 grizzly bears.



For question 1, review the context and ask students what they know from the context. Ask students what key details or information from the problem may be necessary. Help students understand the definition of the variables. Ensure students understand that the values along the x -axis represent the years since 2009. Consider these questions:



- What was the population of grizzly bears in Yellowstone in 2009? In 2019?
- What year does it appear that the population of grizzly bears was at its highest?
- Approximate the population. (~750).



Students may get stuck in the mathematical language in that they lose focus on the real-world context and the language of time. Encourage students to think through their answer to consider what they are saying when they say $G(3)$. Although it is 3 years since 2009, it would be the year 2012. Many students may miss this connection.

1.3.3 Guided Practice: Interpreting Functions (continued)

- f. Explain why there is not a value of t , such that $G(t) = 710.5$.

The value represents the number of grizzly bears. There cannot be a partial grizzly bear, so in this context the value should be a whole number.



- Could evaluating for 710.5 be mathematically possible?
- Why would it not be a value in this scenario?

2. Matthew filled his bathtub, took a bath, and then drained the tub. The function $B(m)$ gives the depth of the water, in inches, m minutes after he began to fill the bathtub.

- a. Explain the meaning of each statement in this context.

$$B(0) = 0$$

At 0 minutes after Matthew began to fill the bathtub, the depth of the water was 0 inches. The tub began with 0 inches of water in it.

$$B(9.5) = B(20)$$

At 9.5 minutes after Matthew began to fill the bathtub, the depth of the water was the same as it was after 20 minutes.

- b. Discuss with your partner why the inequality $B(1) < B(7)$ is likely true in this context.

The depth of the water 1 minute after beginning to fill the bathtub is likely to be less than the depth of water 7 minutes after beginning to fill the bathtub, even if it was already beginning to drain. It is improbable that Matthew would fill the tub, take a bath, and drain it within 7 minutes.



Students may misinterpret the context of question 2 when completing parts a and b. Consider reviewing the context of the problem with students prior to releasing them to work as a method of addressing potential misconceptions ahead of time. Have them reason through the process of water in a bathtub to better have contextual understanding.



For a challenge, have students reverse the inequality symbol and answer the same questions. Would the tub need to be filling or draining for the inequality to be true?

1.3.3 Guided Practice: Interpreting Functions (continued)

3. A biologist is studying the relationship between a tree's diameter and its height. She records the following data for seven different trees using the function $H(d)$, where H represents the height, in feet, and d represents the diameter, in inches.

Diameter (inches)	2	4	6	8	10	12	14
Height (feet)	6	7	10	15	18	22	28

- a. Fill in the blanks in the following sentence to interpret the statement $H(10) = 18$ in this context.

A tree with a diameter of 10 *inches* has a height of 18 *feet*.

- b. Determine the value of $H(14)$.
28 *feet*

- c. Interpret $H(14)$ for the given context.
A tree with a diameter of 14 inches has a height of 28 feet.



Why is it important to connect the labeling of the table with the information from the context?



- *Where on the table did you look to find the value of $H(14)$?*
- *What if the question were written as $H(x) = 14$?*
- *Would the table provide the information needed?*



Allow students to share out not only their answer for question 3 part c, but also to focus on their method for understanding how to interpret the value for the given context. Have students agree or disagree with an interpretation, allowing for student discourse to occur.

1.3.4 Try It: Evaluating and Interpreting Functions

Component Overview: Students will practice being able to evaluate and interpret a function that represents a real-world context. For this activity, release students to work with a partner on completing questions 1 part a and part b. Call the students back together and review the answers as a class.



1.3.4 Try It: Evaluating and Interpreting Functions

1. For groups of 80 or more people, a charter bus company determines the rate per person according to the formula $R(n) = 8 - 0.05(n - 80)$, where n represents the number of people and R represents the rate, in dollars, charged to each customer.

- a. Find each value and interpret it in context.

$$R(75) = 8.25$$

When there are 75 people in the group, the price per customer would be \$8.25. However, we are told that this function is only for groups of 80 people or more so 75 is not a realistic input.

$$R(85) = 7.75$$

When there are 85 people in the group, the price per customer is \$7.75.

- b. Why is $R(75) > R(85)$ in this context?

When the number of people exceeds 80, there is a discount in the price per customer.



Remind students to be careful to review what the information is about, what question is being asked, and what key information can be gathered from the context. Remind students to refer to it when working on interpreting the values and determining reasonableness.



Students may get stumped with the value of $A(75)$, since the problem says for groups 80 or more. Students can evaluate this number but should recognize that the input value itself is not realistic because the company only uses this function when the value is 80 or greater. Do not give this information to students up front but be sure to review this with students at the end of the activity to reinforce the concept of reasonableness.

1.3.5 Your Turn!

Component Overview: Students will independently practice evaluating and interpreting functions that represent a real-world scenario. For this activity, release students to work independently on this activity. Once completed, review answers as a whole group.

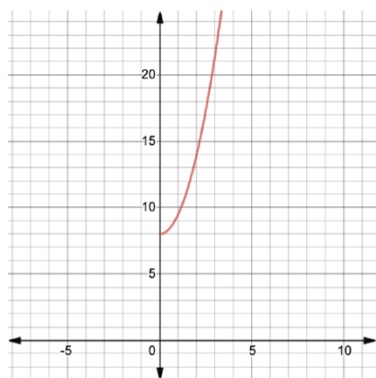


1.3.5 Your Turn!

- The size of a fungal colony is modeled by the equation $z(t) = \frac{3}{2}t^2 + 8$, where z is the size of the colony, in microns, and t is the time, in hours.

A graph of the function $z(t)$ is shown.

Using the graph, approximate the values and interpret their meaning in this context.



- $z(3) = 21$
After 3 hours, the size of the colony is approximately 21 microns.
- $z(0) = 8$
At 0 hours, the size of the colony is 8 microns. The starting size of the colony is 8 microns.
- Why does the graph only display y -values for the function when $x \geq 0$?
In this context, the x -values refer to time. Since time cannot be negative, the corresponding y -values would only appear for x -values that are 0 or greater.



For students who performed poorly on the prior activity, consider using this activity time to work with a small group who may need additional support. Walk them through this problem step-by-step in a gradual release method, consistently checking for understanding.



At the conclusion of this activity, consider having students compare their answers with their partner to determine any potential mistakes that were made. Allow students to share out their answers for question 1 part c to allow for more thoughts and comparisons.



Do you think that a different context would allow for the graph to display y -values for the function when x is less than 0? If yes, what type of context would allow for this to occur?

1.3.6 Wrap-Up: Reflection

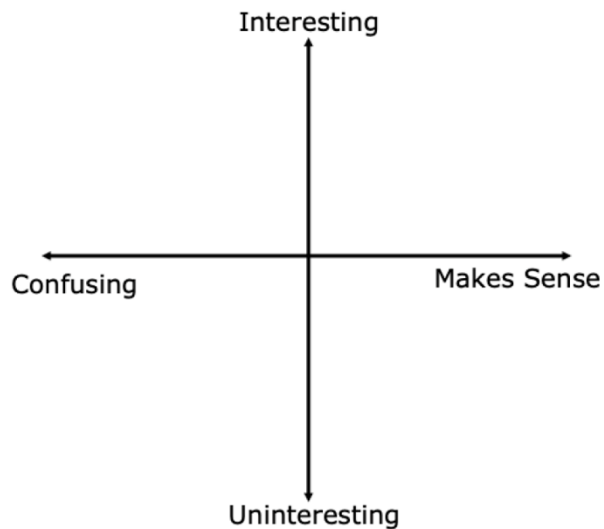
Component Overview: Students will complete a self-assessment of their level of interest and understanding of interpreting functions. This activity should be done independently to allow for accurate student data.



1.3.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you feel you are on today's learning target: "I can evaluate a function for a specific input value of the domain and interpret it in a real-world context."

Answers will vary.



Students may need a demonstration of this to get better data. For example, consider putting this chart on the board with other concepts such as "bread making," and plot a point and use another concept such as "surfing". Highlight for students what their plotted point represents on this graph.

1.4 Applying the Laws of Exponents

Lesson Introduction

 Benchmark	MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.			 Learning Target	- I can apply the Laws of Exponents to write equivalent numerical expressions. - I can evaluate square roots and cube roots.	
 Benchmark Coherence	Building On MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.			Working Toward N/A		
 Essential Questions	- How can the Laws of Exponents help me to evaluate an expression? - What are the key differences between a cube root and a square root?					
 Lesson Narrative	In this lesson, students will revisit the Laws of Exponents, square roots, and cube roots. Students will review the laws of exponents and then transition to working with square and cube roots. They will have the opportunity to practice evaluating expressions using the Laws of Exponents and knowledge of square and cube roots.					
 Component Summary	1.4.1 Warm-Up (~5 min.)	Toothpick Puzzle Challenge (<i>SE p. 23</i>)	1.4.3 Guided Instruction (15 min.)	Extending knowledge of square roots and cube roots (<i>SE p. 24</i>)		
	1.4.2 Refresher (10 min.)	Refresher on the Laws of Exponents (<i>SE p. 23</i>)	1.4.4 Your Turn! (10 min.)	Collaboratively determine the values of square roots and rewriting expressions using the Laws of Exponents (<i>SE p. 25</i>)		
	1.4.5 Wrap-Up (~5 min.)	Cool down that compares two expressions using inequality symbols. (<i>SE p. 26</i>)				
 Resources	Glossary			Materials	Additional Preparation	
	<u>Key Terms:</u> - Principal square root <u>Additional Terms:</u> <table><tr><td> - Cube Root - Square Root - Product of Powers - Quotient of Powers - Power of a Power </td><td> - Power of a Product - Power of a Quotient - Negative Exponents - Identity Exponents - Zero Exponent </td></tr></table>			- Cube Root - Square Root - Product of Powers - Quotient of Powers - Power of a Power	- Power of a Product - Power of a Quotient - Negative Exponents - Identity Exponents - Zero Exponent	N/A
- Cube Root - Square Root - Product of Powers - Quotient of Powers - Power of a Power	- Power of a Product - Power of a Quotient - Negative Exponents - Identity Exponents - Zero Exponent					

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may mistake the definition of a square as double a number. - Students may mistake the definition of a cube as triple a number. - Students may mistake a cube root for a square root.	Students may have varying entry points when accessing this content. Depending on the math courses they have taken prior to Algebra 1, students may have limited working knowledge of exponent laws.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	True or False When presenting the Laws of Exponent table in 1.4.2 Refresher, consider implementing a True or False protocol. Present each answer and ask students if the resulting expression is true or false and allow them to justify why they think that is true or false.	MA.K12.MTR.5.1: Patterns and Structure Prior grade level standards introduced students to the Laws of Exponents. Students will extend that knowledge to include rational exponents. Students can use 1.4.2 Refresher to establish the patterns and structures of the Laws of Exponents.
	Student performance during 1.4.2 Refresher can provide valuable differentiation that may be necessary based on the student's point of entry to the Laws of Exponents. If students struggle to understand, consider differentiating for those students in a later activity.	
	1.4.4 Your Turn! provides a table for students to complete about the Laws of Exponents. Consider creating a tabletop chart or anchor chart that provides references for students to use as they work on this activity.	
	The 1.4.5 Wrap-Up for this activity is a cool down that will provide a formative assessment for students to determine their level of understanding of the content of this lesson. Student performance on this activity can provide data for potential differentiation needs.	
Lesson Notes:		

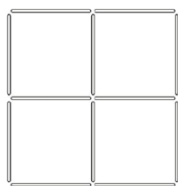
1.4.1 Warm-Up: Toothpick Squares

Component Overview: Students will remove two toothpicks to create two squares with two different sizes. This activity can be done independently.

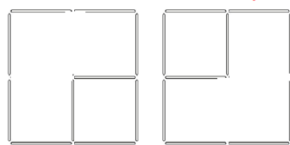


1.4.1 Warm-Up: Toothpick Squares

1. Make two squares of different sizes by removing two toothpicks.



Answers will vary.



Consider providing actual toothpicks. Being able to manipulate the toothpicks themselves, instead of doing it with just the visual from the book, can be a beneficial way for kinesthetic learners to connect to the activity and better complete the task in a way that makes sense to them.



Consider allowing students to share out their methods for solving this problem.

1.4.2 Refresher: Laws of Exponents

Component Overview: Students will receive a refresher instructional lesson on the Laws of Exponents. For this activity, review the exponent law with a variable, explaining what is happening with this law. Then, show students how it applies to integer values and ask students what the equivalent expressions column would look like, providing them the correct answer for their notes. Repeat this process for each exponent law.



1.4.2 Refresher: Laws of Exponents

- Using the given exponent law, complete the table by writing an equivalent exponential expression.

Exponent Law	Expression	Equivalent Expression
Product of Powers $a^m \cdot a^n = a^{m+n}$	$6^5 \cdot 6^2$	6^7
Quotient of Powers $\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^{14}}{5^2}$	5^{12}
Power of a Power $(a^m)^n = a^{m \cdot n}$	$(4^{10})^3$	4^{30}
Power of a Product $(ab)^m = a^m \cdot b^m$	$(5 \cdot 2)^4$	$5^4 \cdot 2^4$
Power of a Quotient $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$	$\left(\frac{3}{11}\right)^2$	$\frac{3^2}{11^2}$
Negative Exponents $a^{-1} = \frac{1}{a}$	9^{-1}	$\frac{1}{9}$
$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$	$\left(\frac{5}{7}\right)^{-1}$	$\frac{7}{5}$
$\left(\frac{a}{b}\right)^{-m} = \frac{b^m}{a^m}$	$\left(\frac{13}{4}\right)^{-3}$	$\frac{4^3}{13^3}$
Identity Exponent $a^1 = a$	572^1	572
Zero Exponent $a^0 = 1$	572^0	1



These laws will be necessary to be successful in the rest of this unit. Consider creating an anchor chart of this information and posting it in a location in the classroom where students can easily reference this as they need throughout the lesson and the unit.



Students may have varying entry points when accessing this content. Depending on the math courses they have taken prior to Algebra, students may have limited working knowledge of these exponent laws. Therefore, this activity will be a refresher for some and new content for others.



How can knowledge of these exponent laws help simplify the evaluation process?

1.4.3 Guided Instruction: Building on Square Roots and Cube Roots

Component Overview: Students will receive guided instruction on expanding their knowledge of square roots and cube roots. For this activity, walk students through the instruction provided in this section. Consider stopping whenever an answer is required to elicit student response, but ensuring they have the correct answer for their notes.



1.4.3 Guided Instruction: Building on Square Roots and Cube Roots

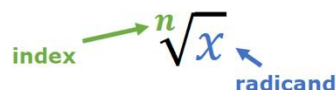
To square a number means to

- multiply it by itself.
- multiply it by 2.

$$7^2 = 7 \cdot 7 = 49$$

$$(-7)^2 = (-7) \cdot (-7) = 49$$

The inverse of squaring a number is taking the square root ($\sqrt{\quad}$).



To evaluate $\sqrt[3]{25}$, also written as $\sqrt[3]{25}$, ask, "What times itself equals 25?"

$$5 \cdot 5 = 25 \text{ or } -5 \cdot -5 = 25$$

Therefore, $\sqrt{25}$ is 5 and -5 .



All radicands greater than or equal to 0, $x \geq 0$, have two square roots, a positive and a negative. The positive square root is referred to as the **principal square root**.



- How can the value of a number squared differ if the understanding of what the square means is "multiply it by 2" instead of "multiply it by itself"?
- How might 2^2 be confusing when discussing the definition?



Consider using this example as a potential anchor chart or visual reference for students to recall. While they may not fluently use these terms in this lesson, they will need to use them in future lessons. Use these terms when doing a think-aloud or providing instruction to reinforce the terms.

For example, say "index of 3" when describing "cube roots" so students can continue to make the connections of a cube root and its index value of 3.



Consider using a language activity for this section. For example, write $\sqrt{25}$ on the board. For each student, ask for one piece of information about the radical. This can be the index, a square root, the principal square root, or what it means. Ask one student for one, another for another, and repeat for all students. You will repeat what you're asking for, but it will help students connect the vocabulary to the content.

1.4.3 Guided Instruction: Building on Square Roots and Cube Roots (continued)

The square root of 16, written as $\sqrt{16}$, is 4 and -4 because both 4^2 and $(-4)^2$ are equal to 16.

To cube a number means to

- multiply the number by 3.
- triple the number.
- multiply the number by itself three times.

$$6^3 = 6 \cdot 6 \cdot 6 = 216 \quad (-6)^3 = (-6) \cdot (-6) \cdot (-6) = -216$$

The cube root of a number is a value that when multiplied by itself three times gives that number.

$$27 = 3^3 \text{ so } \sqrt[3]{27} = 3$$

$$-27 = (-3)^3 \text{ so } \sqrt[3]{-27} = -3$$

The cube root of 64, written as $\sqrt[3]{64}$, is 4 because 4^3 is 64.

$$4 \cdot 4 \cdot 4 = 64$$



- How can the value of a number cubed differ if the understanding of what the cube means is “multiply it by 3” or “triple the number” instead of “multiply it by itself”?
- How might 3^3 be confusing when discussing the definition?



- Why do you think the result of a negative number squared is positive, but when it's cubed it would be negative?
- What if a number was raised to the 4^{th} or 9^{th} power?



For a challenge, have students determine a number that would be the same when squared as it is when cube rooted. Have them share how they arrived at that answer and if changing the sign of that number would cause it to no longer be true.

1.4.4 Your Turn!

Component Overview: Students will work on writing expressions using the Laws of Exponents. For this activity, begin by releasing students to do question 1 on their own, and then bring the class back together to review the answers. Consider having students work with their partner to complete the table in question 2, then bring the class back together to review the answers. Then, students will work independently on questions 3 and 4.



1.4.4 Your Turn!

1. Find the value(s) of each.

a. $\sqrt{121} = \pm 11$

b. $\sqrt[3]{125} = 5$

2. Work with a partner to complete the chart.

Expression	Exponent Law	Equivalent Expression
$8^2 \cdot 8^4$	Product of Powers	8^6
$10^4 \cdot 10^3$	Product of Powers	10^7
$(7^5)^3$	Power of a Power	7^{15}
$(4^8)^4$	Power of a Power	4^{32}
$\frac{2^{12}}{2^{10}}$	Quotient of Powers	2^2
$\frac{6^{20}}{6^{11}}$	Quotient of Powers	6^9
$(3 \cdot 7)^5$	Power of a Product	$3^5 \cdot 7^5$
$(13 \cdot 2)^5$	Power of a Product	$13^5 \cdot 2^5$
$\left(\frac{1}{7}\right)^4$	Power of a Quotient	$\frac{1^4}{7^4}$
$\left(\frac{6}{25}\right)^3$	Power of a Quotient	$\frac{6^3}{25^3}$
$\left(\frac{2}{5}\right)^{-2}$	Negative Exponents	$\frac{5^2}{2^2}$
$11^3 \cdot 11^{-5}$	Product of Powers	$\frac{1}{11^2}$
45^{-1}	Negative Exponents	$\frac{1}{45}$



For question 1, consider having students find the value without a calculator. The square root of question 1 part a, and the cube root of question 1 part b can be done without a calculator. Students may provide both roots for the answer to question 1 part a. Use this to help establish that both roots are correct, but that in the future the principal square root should be considered the answer.



Be strategic on partners and grouping for question 2. Some students will have a strong background knowledge, having seen these exponent laws in a prior course, and some may never have been exposed to these in their course progression. If you have a class full of 8th grade students with limited exposure, consider doing a gradual release into question 2.



Watch for student performance on both sides of the rows. Students may understand how to evaluate the expression using the law, but may not understand how to create the expression from the final value.

Ensure that students understand what an equivalent expression is compared to the value of the expression.

1.4.4 Your Turn! (continued)

3. Marques, Pablo, and Tandice used the product of powers law to write an equivalent expression for $5^{-7} \cdot 5^6$. Each student's equivalent expression is shown in the table.

Marques	Pablo	Tandice
$\frac{1}{5}$	-5	5^{-1}

- a. Which student's answer is not an equivalent expression for $5^{-7} \cdot 5^6$?
Pablo
- b. Which student also applied the negative exponent law?
Marques

4. Find the value of the variable(s) that makes the equation true.

a. $7^{10} = 7^5 \cdot 7^f$ b. $\sqrt[3]{h} \cdot \sqrt[3]{h} \cdot \sqrt[3]{h} = -13$ c. $\sqrt{m} \cdot \sqrt{m} = 26$

$f = 5$ $h = -13$ $m = 26$

d. $\frac{7^5}{6^5} = \left(\frac{6}{7}\right)^y$ e. $35^{12} = (35^x)^2$ f. $\left(\frac{9}{10}\right)^{-1} = \frac{c}{r}$

$y = -5$ $x = 6$ $c = 10$

$r = 9$

g. $18^0 = \frac{18^w}{18^7}$

$w = 7$



- Which exponent law was the easiest for you to recall and use?
- Which exponent law was the most difficult for you to use?



Allow students to share out their answers to question 3 parts a and b. Engage students in a polling protocol where a student pair is asked to share out their answer to the question, and then other students either agree or disagree with the answer and provide justification.



Consider having students articulate and explain how each student arrived at their answer.



For sharing out the answers to question 4, consider infusing physical movement into the activity. For example, post each of the seven expressions at different parts of the room or on different spaces on the board. Group students into seven groups and have each group show their steps on the board (or poster paper) and share with the class how they determined the value, what law they used, and any strategies they used.

1.4.5 Wrap-Up: Cool Down

Component Overview: Students will complete a cool down that has students compare two expressions using inequality symbols. This activity is designed to be done independently to provide data on how each student is doing related to mastery of the content of this lesson.



1.4.5 Wrap-Up: Cool Down

1. Use the symbols $<$, $>$, or $=$ to compare each pair of values. Then, explain your choice.

Expression	Explanation
$2^3 > 2 \cdot 3$	$2^3 = 2 \cdot 2 \cdot 2 = 8$ $2 \cdot 3 = 6$ $8 > 6$
$5^2 < 2^5$	$5^2 = 5 \cdot 5 = 25$ $2^5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$ $25 < 32$
$100^1 > 1^{100}$	$100^1 = 100$ $1^{100} = 1$ $100 > 1$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

1.5 Exploring Rational Exponents

Lesson Introduction

 Benchmark	MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.		 Learning Target	- I can apply the Laws of Exponents to rational exponents. - I can convert between expressions involving rational exponents and expressions involving radicals.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.		N/A	
 Essential Questions	- How can a number with a rational number exponent represent the same value in radical form? - What is the relationship between a number with a rational exponent and a number in radical form?			
 Lesson Narrative	In this lesson, students will continue their learning with the Laws of Exponents to focus on generating equivalent numerical expressions. Students will explore the relationship between rational number exponents and radical form. They will practice how to rewrite expressions between forms and evaluate them.			
 Component Summary	1.5.1 Warm-Up (~5 min.)	Bell Work on the Laws of Exponents (<i>SE p. 27</i>)	1.5.4 Exploration (5 min.)	Explore roots other than square and cube with translating between rational exponent and radical expression (<i>SE p. 30</i>)
	1.5.2 Exploration (10 min.)	Explore generating equivalent expressions and evaluating rational exponents (<i>SE p. 28</i>)	1.5.5 Guided Instruction (10 min.)	Instruction on applying exponent laws with rational exponents (<i>SE p. 30</i>)
	1.5.3 Guided Instruction (10 min.)	Instruction on understanding the relationship between rational exponents and radical expressions (<i>SE p. 29</i>)	1.5.6 Try It (5 min.)	Practice rewriting expressions using the Laws of Exponents (<i>SE p. 31</i>)
	1.5.7 Wrap-Up (~5 min.)	Ticket Out the Door to write an equivalent expression using the Laws of Exponents. (<i>SE p. 32</i>)		
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Rational Exponent - Radical Form - Equivalent Expression		Work through each of the explorations prior to having students try these. They are designed to lead students to a discovery in a very specific way.	
		Materials		
		Calculator for 1.5.1 Warm-up		

 Teacher Tips	Common Misconceptions - Students may not make the connection between the two forms. - Students may put the numerator of the fraction in the exponent as the index of the radical form. - Students may make calculation errors when determining the value of the expression.	Things to Consider - This concept is new to most students and may be abstract in their thinking. It's important to avoid any tricks with where values go and focus on the actual mechanics of why the two forms (radical and rational exponent) are equivalent. - If time is an issue, then consider assigning 1.5.6 Try It for homework.
	Literacy and Language Routines Think-Pair-Share During 1.5.5 Guided Instruction and 1.5.6 Try It, students are given the opportunity to try a question on their own. Consider leveraging the Think-Pair-Share protocol to allow students to work with a partner after they've done it on their own and discuss their methods and findings.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Students have previously worked with Laws of Exponents and square and cube roots. This lesson prompts students to apply that knowledge to establish that the Laws of Exponents apply to rational exponents.
	 Differentiate Instruction	
Lesson Notes:		

1.5.1 Warm-Up: Bell Work

Component Overview: Students will complete a Bell Work activity, applying what they learned about Laws of Exponents from the prior lesson to generate equivalent expressions and evaluate to find the value. This activity is designed to be done independently.



1.5.1 Warm-Up: Bell Work

- Use the laws of exponents to write an equivalent expression. Then, give the value of each expression. If necessary, use a calculator to find the value of the equivalent expression.

$(8^{-5})(8^3)$ Equivalent Expression $\frac{1}{8^2}$ Value $\frac{1}{64}$	$(3^4)^6$ Equivalent Expression 3^{24} Value 282,429,536,481
$\left(\frac{2}{11}\right)^3$ Equivalent Expression $\frac{2^3}{11^3}$ Value $\frac{8}{1331}$	$\left(\frac{12}{36}\right)^{-4}$ Equivalent Expression $\frac{36^4}{12^4}$ Value 81



For this Warm-Up, consider allowing students to use a calculator to find the value of these expressions. While the first one is appropriate without a calculator, students may spend an excessive amount of time trying to find the value of 3^{24} instead of moving onto the next questions in this activity.

It is possible that students give equivalent expressions that are not in the form of the answer key. Allow the students the freedom of writing an equivalent expression in a variety of ways. If students provide a variety of responses, consider sharing the variety with the class to expand all students' understanding of equivalent expressions.



If time allows, consider asking the following while reviewing the answers:

- What Law of Exponents did you use to generate the equivalent expression?
- How did you know you could use the law you chose?
- How did using the Law of Exponents help you determine the value?

1.5.2 Exploration: Rational Exponents

Component Overview: Students will work with a partner to conduct an exploration with generating equivalent expressions and evaluating rational exponents. For this activity, begin by releasing students to complete question 1 through 3 with their partner and call the class back together to discuss their answers. Repeat this process for questions 4 and 5, 6 – 9, and 10 – 13.



1.5.2 Exploration: Rational Exponents

- Fill in the blank with the value of each.

$$16^0 = 1$$

$$16^1 = 16$$

- Using the values above, guess the value of $16^{\frac{1}{2}}$.

Answers will vary. I guess that it would be somewhere between 1 and 16. I guess that it would be 8, because that's half of 16.

- Explain the strategy you used.

Answers will vary. I found out half of the value when raised to the power of 1. I decided to use a value that was halfway to 16.

- Find the value of $16^{\frac{1}{2}}$ using a calculator.

4

- Discuss with your partner what you notice about the relationship between 16 and the value of $16^{\frac{1}{2}}$.

Answers will vary. The value is the same as the square root of 16.

- Predict the value of $49^{\frac{1}{2}}$. Check using your calculator.

7

- Predict the value of $100^{\frac{1}{2}}$. Check using your calculator.

10

- Write an expression equivalent to $a^{\frac{1}{2}}$.

$\sqrt{a}$



What Law of Exponents did you use to determine the value of 16^0 ?



After questions 1 – 3, consider having students share out the different guesses that they came up with for the value in question 2. Poll the class to determine who came up with that same value, charting each guess on the board until all guesses have been accounted for. Have students also share their strategy for determining their guesses to allow other students to hear different approaches to the problem.



After questions 4 and 5, consider using an on-screen calculator, document camera, or other method to show students how to input a fractional exponent into a calculator. Students may struggle with this skill.

1.5.2 Exploration: Rational Exponents (continued)

9. Check your answer with your partner. If necessary, update your answer.

Answers may vary. Students should arrive at correct answer after collaboration.

10. Predict the value of $8^{\frac{1}{3}}$. Check using your calculator.
2

11. Predict the value of $125^{\frac{1}{3}}$. Check using your calculator.
5

12. Write an expression equivalent to $a^{\frac{1}{3}}$.
 $\sqrt[3]{a}$

13. Check your answer with your partner. If necessary, update your answer.

Answers may vary. Students should arrive at correct answer after collaboration.



After questions 6 – 9, engage students in a discussion about what they uncovered in question 8. Have students explain how they chose the answer that they did from the work they did in questions 5 – 7.



- *What relationship did you notice between the values in the rational exponent and the values in the radical form?*
- *Do you think those patterns would hold true for any value outside of 2 and 3?*

1.5.3 Guided Instruction: Why Does That Work?

Component Overview: Students will receive guided instruction on understanding the relationship between rational exponents and radical expressions.



1.5.3 Guided Instruction: Why Does That Work?

1. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}}$	$6^{\frac{1}{2} + \frac{1}{2}} = 6^1$	6

2. What is the whole number that completes this equation?

$$\sqrt{6} \cdot \sqrt{6} = 6$$

Because $6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}} = 6$ and $\sqrt{6} \cdot \sqrt{6} = 6$, then $6^{\frac{1}{2}}$ is equivalent to $\sqrt{6}$.

3. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}}$	$5^{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}} = 5^1$	5

4. What is the whole number that completes the equation?

$$\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = 5$$

Because $5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} = 5$ and $\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = 5$, then $5^{\frac{1}{3}}$ is equivalent to $\sqrt[3]{5}$.

5. Rewrite each exponential expression as a radical expression. Then, evaluate each expression.

Exponential Expression	Radical Expression	Value
$81^{\frac{1}{2}}$	$\sqrt{81}$	9
$8^{\frac{1}{3}}$	$\sqrt[3]{8}$	2
$(-216)^{\frac{1}{3}}$	$\sqrt[3]{-216}$	-6



- What is the relationship between the square root of 6 and 6 raised to the one-half power?
- Do you think this relationship would exist with any values or is it strictly for the square roots? Why or why not?



After questions 1 and 2, engage students in a discussion related to the connection between square roots and values raised to the one-half power. After questions 3 and 4, rekindle the discussion of the relationship between cube roots and values raised to the one-third power. Draw student attention to the process connection between the two different types.



Challenge students to create an additional row, from scratch, to add to the table in question 5. Have them complete all three columns of their new row.

1.5.4 Exploration: Other Roots

Component Overview: Students will explore roots other than square and cube with translating between rational exponent and radical expression. For this activity, review the narrative language prior to the table. Model the first row for students and then release students to work with a partner on the remaining rows.



1.5.4 Exploration: Other Roots

Rewriting exponential expressions as radical expressions does not only apply to square roots and cube roots. Other roots apply these same principals.

Complete the table to explore other rational exponents and roots.

Rational Exponent	Radical Expression	Value
$81^{\frac{1}{4}}$	$\sqrt[4]{81}$	± 3
$256^{\frac{1}{4}}$	$\sqrt[4]{256}$	± 4
$(-32)^{\frac{1}{5}}$	$\sqrt[5]{-32}$	-2
$(1,000,000,000)^{\frac{1}{9}}$	$\sqrt[9]{1,000,000,000}$	10



Relate what students have done with cube and square roots in earlier activities to situations where the roots are different. Remind students to refer to the work they just completed in the prior activity as a basis for consideration when completing this exploration. When the activity is complete, have students share out their answers and strategies of how they thought through each question.



- What part of this process was similar to what you did with cube and square roots in the prior activity?
- What parts were different?



Challenge students to create an additional row, from scratch, to add to the table in this question. Have them complete all three columns of their new row.

1.5.5 Guided Instruction: Exponent Laws With Rational Exponents

Component Overview: Students will receive guided instruction on applying exponent laws with rational exponents.



1.5.5 Guided Instruction: Exponent Laws with Rational Exponents

1. Kalini's teacher asked her to rewrite $\left((100)^{\frac{1}{2}}\right)^3$ as an equivalent exponential expression. Kalini says that she can use a law of exponents to multiply the powers.
 - a. Which law of exponents is Kalini using?
Power of a Power
 - b. Using the laws of exponents, what is an equivalent exponential expression for $\left((100)^{\frac{1}{2}}\right)^3$?
 $100^{\frac{3}{2}}$
2. The expression $27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}}$ can also be rewritten using the laws of exponents.
 - a. Rewrite the expression.
 $27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}} = 27^2$
 - b. Which exponent law was used?
Product of Powers
 - c. Check your answer with your partner.
Answers may vary. Students should end with correct answer after collaboration.
 - d. Evaluate the exponential expression.
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During the modelling of these questions, allow for students to share what they think about each part of the question, referring to the context. Model for students a way of thinking through each part of the question, using appropriate mathematical language to help reinforce what is happening at each step.



Consider writing each step of the process for questions such as 1b and 2a. For each step, use a different colored marker to label what action was taken at each step in a proof-like fashion. This way, students will be able to visualize each step and have a reference for future questions of a similar nature.



For a challenge, consider having students create a problem using a different rational exponent being rewritten by a different Law of Exponents.

1.5.5 Guided Instruction: Exponent Laws With Rational Exponents (continued)

3. Maeko explained his work for rewriting the following expression.

Explanation	$16^{-\frac{8}{2}} \cdot 16^{\frac{1}{2}}$
I first used the product of powers to rewrite the expression.	$16^{-\frac{8}{2} \cdot \frac{1}{2}} = 16^{-\frac{8}{4}}$
Next, I simplified the fractional exponent.	4^{-2}
Finally, I rewrote the expression using the negative exponent law.	$\frac{1}{16}$

Check Maeko's work and explain any errors he may have made.

Maeko made mistakes in the first and second step. In the first step, he multiplied the exponents instead of adding them. When he rewrote the second step, he changed the base to 4.



Think-Pair-Share

Have students complete this question on their own first. Then, allow each student to compare their work with a partner to discuss their approach and final answer. Together, each pair should come to a consensus on the correct answer. Have pairs share out their answer and determine what other pairs came to the same conclusion, repeating until all answers have been heard.



- What was Maeko attempting to do when he made the mistake? What went wrong in his process?
- What advice would you give him to help explain what went wrong and how he can avoid making that mistake again in the future?

1.5.6 Try It: Writing Rational Exponents

Component Overview: Students will work independently to practice rewriting expressions using the Laws of Exponents then evaluate the expression.



1.5.6 Try It: Writing Rational Exponents

Rewrite each expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$5^{\frac{26}{7}} \cdot 5^{\frac{9}{7}}$	5^5	3,125
$\frac{23^{\frac{8}{3}}}{23^{\frac{2}{3}}}$	23^2	529
$125^{\frac{7}{15}} \cdot 125^{-\frac{2}{15}}$	$125^{\frac{1}{3}}$	5
$169^{-\frac{5}{22}} \cdot 169^{\frac{25}{22}} \cdot 169^{-\frac{31}{22}}$	$169^{-\frac{1}{2}}$	$\frac{1}{13}$



Challenge students to create an additional row, from scratch, to add to the table in question 5. Have them complete all three columns of their new row.

1.5.7 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses students' ability to write an equivalent expression using the Laws of Exponents. This activity should be done independently to allow for accurate data on student mastery of the content of this lesson. Students should not use a calculator to complete the Wrap-Up.



1.5.7 Wrap-Up: Ticket Out the Door

1. Find the value of the variable.

a. $\sqrt{14} = 14^{\frac{1}{k}}$

$k = 2$

b. $50^{\frac{1}{3}} = \sqrt[n]{50}$

$n = 3$

2. Complete the table by writing an equivalent expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$6^{-\frac{1}{6}} \cdot 6^{\frac{19}{6}}$	6^3	216
$\frac{7^{\frac{1}{2}}}{7^{\frac{5}{2}}}$	7^{-2}	$\frac{1}{49}$



Student performance on this Wrap-Up can provide data that indicates additional support and differentiation that may be necessary. The following can be determined:

- Question 1
 - Part A: Rewriting radical form to rational exponent form
 - Part B: Rewriting rational exponent form to radical form
- Question 2
 - Equivalent Expression
 - Calculating Actual Value

1.6 Writing Equivalent Expressions Using Rational Exponents

Lesson Introduction

 Benchmark	MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.		 Learning Targets	- I can apply the Laws of Exponents to rational exponents. - I can convert between expressions involving rational exponents and expressions involving radicals.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.		N/A	
 Essential Questions	- How can I use the relationship between rational exponents and radical expressions to convert between the two forms? - How can the Laws of Exponents be used to convert between rational exponent form and radical expression form?			
 Lesson Narrative	In this lesson, students will continue using the Laws of Exponents to convert between rational exponent form and radical expression form. Students will receive a greater amount of exploration and practice in this lesson, giving them the opportunity to reinforce what they have been learning in the last several lessons and to improve their fluency.			
 Component Summary	1.6.1 Warm-Up (~10 min.)	Identify equivalent expressions matching activity (<i>SE p. 33</i>)	1.6.4 Activity (5-10 min.)	Card sort activity with expressions written in rational exponent form and radical expression form into equivalent sets (<i>SE p. 34</i>)
	1.6.2 Guided Practice (5 min.)	Writing equivalent expressions practice (<i>SE p. 33</i>)	1.6.5 Try It (5-10 min.)	Practice writing equivalent expressions in rational exponent and radical expression form with a partner (<i>SE p. 35</i>)
	1.6.3 Exploration (5-8 min.)	Match expressions in rational exponent form to radical expression form (<i>SE p. 34</i>)	1.6.6 Your Turn! (5 min.)	Practice rewriting radical expressions into rational exponent form independently (<i>SE p. 35</i>)
	1.6.7 Wrap-Up (~5 min.)	Cool down that covers equivalent expressions in both radical and rational exponent form (<i>SE p. 36</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A		- Cards for Card Sort Activity - Calculators	Work out each of the activities on your own prior to planning for the lesson. This will help you get an understanding of the activities to best pace your lesson accordingly.
	<u>Additional Terms:</u> - Rational Exponent - Radical Form - Equivalent Expression			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not make the connection between the two forms. - Students may put the numerator of the fraction in the exponent as the index of the radical form. - Students may make calculation errors when determining the value of the expression.	- This lesson has a lot of collaborative activities. Consider your student learning styles and specific classroom needs when pacing out the activities and discussions. - Consider your students to determine if having a calculator for 1.6.1 Warm-Up will help students determine equivalent expressions. The focus should be on relating the equivalent expressions, and for some students the different bases may be a struggle for them. - Calculators are used throughout the lesson to help support students learning equivalent rational exponents and radical expressions. - If time is an issue, then consider assigning 1.6.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share During the collaborative activities of this lesson, students can engage in a Think-Pair-Share protocol. Consider leveraging the Think-Pair-Share protocol to allow students to work with a partner after they've done it on their own and discuss their methods and findings.	MA.K12.MTR.4.1: Discussion and Reflection Students will work collaboratively for most of this lesson. The collaboration will require students to engage in discussions that require them to reflect on their own thinking and the thinking of others.
 Differentiate Instruction	Student performance on 1.6.5 Try It will provide data on where students are struggling to be able to provide them differentiated and targeted support during 1.6.6 Your Turn!. For 1.6.4 Activity, consider creating physical cards for the activity to allow kinesthetic learners to have a more hands-on experience with this activity. Students will complete a Ticket Out the Door that assesses students' understanding of equivalent expressions in radical expression in rational exponent form.	
Lesson Notes:		

1.6.1 Warm-Up: Identifying Equivalent Expressions

Component Overview: Students will apply prior knowledge to identify equivalent expressions. This activity is designed to have students complete question 1 on their own and then work with a partner to complete questions 2 and 3.



1.6.1 Warm-Up: Identifying Equivalent Expressions

Three lists of numbers are shown.

List A	List B	List C
$8 \cdot 8 \cdot 8$	2^9	24
$4 \cdot 16 \cdot 2$	$6 \cdot 4$	$\frac{1}{32}$
$2 \cdot 16$	$\left(\frac{1}{2}\right)^5$	512
$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}$	$2^2 \cdot 2^3$	128
$4 + 4 + 4 + 4 + 4 + 4$	$2 \cdot 4^3$	32



Students may need additional support with the directions of this activity. Consider completing one to show students what to do or provide them explicit instructions on what to do on their own and how many matches each list should have and not have.

1. Choose an expression from List A and match it with an equivalent expression from List B and List C.

$8 \cdot 8 \cdot 8$	2^9	512
$4 \cdot 16 \cdot 2$	$2 \cdot 4^3$	128
$2 \cdot 16$	$2^2 \cdot 2^3$	32
$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}$	$\left(\frac{1}{2}\right)^5$	$\frac{1}{32}$
$4 + 4 + 4 + 4 + 4 + 4$	$6 \cdot 4$	24

2. Explain to your partner how you know the expressions you matched are equivalent.

Answers will vary.

3. Listen to your partner explain the match they found. If you disagree, discuss your thinking and work to reach an agreement.

Answers will vary.

1.6.2 Guided Practice: Writing Equivalent Expressions

Component Overview: Students will practice writing equivalent expressions through a structured format using the Laws of Exponents.



1.6.2 Guided Practice: Writing Equivalent Expressions

- Use the Power of a Power Law of Exponents to write an equivalent exponential expression for the following two expressions.

a. Expression 1: $\left(a^{\frac{1}{c}}\right)^b = a^{\left(\frac{1}{c} \cdot b\right)} = a^{\frac{b}{c}}$

b. Expression 2: $(a^b)^{\frac{1}{c}} = a^{(b \cdot \frac{1}{c})} = a^{\frac{b}{c}}$

- c. Explain why the expressions $\left(a^{\frac{1}{c}}\right)^b$ and $(a^b)^{\frac{1}{c}}$ are equivalent.

Answers will vary. They are equivalent because when you multiply the exponents together, they result in the same final value. They are equivalent because the numerator is 1, when they are multiplied, the values are the same.

- Two students evaluated an expression using two different methods.

Jerry	Antonio
$\left((81)^{\frac{1}{2}}\right)^3$	$\left((81)^{\frac{1}{2}}\right)^3$
$(\sqrt{81})^3$	$81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}}$
$(9)^3$	$\sqrt{81} \cdot \sqrt{81} \cdot \sqrt{81}$
729	$9 \cdot 9 \cdot 9$
	729

Explain the difference between the two methods.

Jerry converted from exponent form to radical form, simplified, and then evaluated. Antonio rewrote the power of a power rule in expanded form, then converted from exponent to radical form, and then evaluated.



After question 1 part c, allow students to share out their thinking around why the expressions are equivalent. Consider choosing different input values to show these expressions being evaluated with a numeric value to show their equivalency.



- How do you think it's possible to be able to evaluate the same expression using two different methods and still get the same answer?
- How does the relationship between rational exponents and radical form allow that to happen for Jerry and Antonio?

1.6.3 Exploration: Matching Expressions

Component Overview: Students will work with a partner to match expressions in rational exponent form to radical expression form. For this activity, introduce the directions for students so they are aware of what they are required to do. Release students to complete questions 1 – 3 and review all answers together as a class.



1.6.3 Exploration: Matching Expressions

1. Use your calculator to help you match each expression in the first table with an equivalent expression in the second table. Record your answers in the table below.

Rational Exponent	Radical Expression
$(64)^{\frac{3}{2}}$	$\sqrt[4]{64}$
$(64)^{-\frac{1}{2}}$	$\sqrt[3]{64^2}$
$(64)^{\frac{2}{3}}$	$(\sqrt[3]{64})^4$
$(64)^{\frac{1}{4}}$	$\sqrt{64}$
$(64)^{\frac{1}{2}}$	$(\sqrt{64})^3$
$(64)^{\frac{4}{3}}$	$\frac{1}{\sqrt{64}}$

Rational Exponent	Radical Expression	Rational Exponent	Radical Expression
$(64)^{\frac{3}{2}}$	$(\sqrt{64})^3$	$(64)^{\frac{1}{4}}$	$\sqrt[4]{64}$
$(64)^{-\frac{1}{2}}$	$\frac{1}{\sqrt{64}}$	$(64)^{\frac{1}{2}}$	$\sqrt{64}$
$(64)^{\frac{2}{3}}$	$\sqrt[3]{64^2}$	$(64)^{\frac{4}{3}}$	$(\sqrt[3]{64})^4$

2. Discuss with your partner the relationship between the rational exponent and the radical expression.
Answers will vary. I think it looks like the denominator of the exponent is the same as the index on the radical. I think the negative exponent rule still applies in both situations.
3. Work with your partner to complete the following statement.

$$\left(a^{\frac{1}{c}}\right)^b = (a^b)^{\frac{1}{c}} = (\sqrt[c]{a})^b = \sqrt[c]{(a^b)}$$



Teaching one method of relating rational exponents to an equivalent radical expression does not give the student the autonomy of choosing a method that makes sense to them. The flexibility that students gain from knowing that there are multiple ways of representing rational expressions and radical expressions builds fluency.



Question 2 is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.



Spend a little extra time on question 3. In theory, students will have been able to discover this pattern through this activity. Use the time for reviewing the activity and answers to make sure that misconceptions are addressed and their statement for question 3 is correct.

1.6.4 Activity: Card Sort

Component Overview: Students will work with a partner to sort cards with expressions written in rational exponent form and radical expression form into equivalent sets. For this activity, provide students with instructions on what they will be doing and release students to work with their partner to pair them into equivalent pairs.



1.6.4 Activity: Card Sort

1. Work with your partner to sort the cards into sets of equivalent expressions.

$(18^2)^{\frac{1}{3}}$	$(21^{\frac{1}{4}})^5$	$18^{\frac{3}{2}}$	$(21^5)^{\frac{1}{4}}$
$(\frac{1}{21^4})^{\frac{1}{5}}$	$\frac{1}{(\sqrt[4]{21})^{-5}}$	$21^{\frac{5}{4}}$	$(18^3)^{\frac{1}{2}}$
$(18^{\frac{1}{2}})^3$	$(21^{\frac{1}{5}})^{-4}$	$18^{\frac{2}{3}}$	$\frac{1}{(\sqrt[5]{21})^4}$
$21^{-\frac{4}{5}}$	$(\sqrt[3]{18})^2$	$(\sqrt{18})^3$	$(18^{\frac{1}{3}})^2$

Record your matches below.

Equivalent Expression
$(18^2)^{\frac{1}{3}}$ $18^{\frac{2}{3}}$ $(\sqrt[3]{18})^2$ $(18^{\frac{1}{3}})^2$
$(21^{\frac{1}{4}})^5$ $(21^5)^{\frac{1}{4}}$ $21^{\frac{5}{4}}$ $\frac{1}{(\sqrt[4]{21})^{-5}}$
$18^{\frac{3}{2}}$ $(18^3)^{\frac{1}{2}}$ $(18^{\frac{1}{2}})^3$ $(\sqrt{18})^3$
$(\frac{1}{21^4})^{\frac{1}{5}}$ $(21^{\frac{1}{5}})^{-4}$ $\frac{1}{(\sqrt[5]{21})^4}$ $21^{-\frac{4}{5}}$



This activity can be done with the material written in the book by using lists, different markings, or coloring materials to indicate the matching sets. For kinesthetic learners, consider creating actual cards for them to physically compare and match while working on this activity. This can be done by printing the cards and cutting them out or writing each expression onto an index card.



For the whole-group share-out, consider a rapid whiteboard activity. Have a student write one of the expressions and give the rest of the class 30 seconds to write the correct matching expression and hold them up for everyone to see. Discuss the correct answer and repeat this process until all expressions have been completed.



For a challenge, consider having students create expression cards and allowing them time to give those cards to another classmate for them to sort.

1.6.5 Try It: Writing Equivalent Expressions

Component Overview: Students will practice writing equivalent expressions as a rational exponent and radical expression with a partner.



1.6.5 Try It: Writing Equivalent Expressions

1. Complete the table.

$(a)^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
$(100)^{\frac{3}{2}}$	$\left((100)^{\frac{1}{2}}\right)^3$	$((100)^3)^{\frac{1}{2}}$	$(\sqrt{100})^3$	$\sqrt{100^3}$
$60^{\frac{5}{4}}$	$\left((60)^{\frac{1}{4}}\right)^5$	$((60)^5)^{\frac{1}{4}}$	$(\sqrt[4]{60})^5$	$\sqrt[4]{60^5}$
$8^{\frac{3}{5}}$	$\left((8)^{\frac{1}{5}}\right)^3$	$((8)^3)^{\frac{1}{5}}$	$(\sqrt[5]{8})^3$	$\sqrt[5]{8^3}$
$27^{\frac{2}{3}}$	$\left((27)^{\frac{1}{3}}\right)^2$	$((27)^2)^{\frac{1}{3}}$	$(\sqrt[3]{27})^2$	$\sqrt[3]{27^2}$

2. Based on the equivalent expressions in the table, write three equivalent expressions for $(\sqrt{17})^8$.

$$\sqrt{17^8}; 17^{\frac{8}{2}}; \left((17^{\frac{1}{2}})^8\right)$$



Students may need additional support with getting started on the table activity in question 1. Begin by showing students an example or two from the first row. Students may also struggle with how to work backwards in the remaining row. Show students how to work backwards in the second row similarly, so they can better understand how to work through the table.



Student performance on this activity will provide data on which equivalent representation, if any, students struggle with. Consider making notes of where students are struggling to be able to provide them differentiated and targeted support during 1.6.6 Your Turn! or in later lessons. This Try It is helping students extend their fluency of the Laws of Exponents.

1.6.6 Your Turn!

Component Overview: Students will work independently to practice rewriting radical expressions into rational exponent form. For this activity, release students to complete question 1 on their own, and review the answers as a class.



1.6.6 Your Turn!

1. Rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$.

$$\frac{(\sqrt[3]{72})^5}{72^{\frac{5}{3}}}$$

$$\frac{\sqrt{5^3}}{5^{\frac{3}{2}}}$$

$$\frac{1}{(\sqrt[3]{20})^4}$$

$$\frac{1}{\sqrt{6}}$$



Students who struggled in the prior activity could use this time to get additional support. Consider pulling a small group together to review these questions and use the gradual release model to walk students through the process again in a smaller group. These students may not complete all four of the questions in the allotted time, but their goal will be to understand what they didn't in prior instruction and activities.



For a challenge, consider having students create a problem and trading with another student to have them rewrite it into the other form.

1.6.7 Wrap-Up: Cool Down

Component Overview: Students will complete a cool down that assesses their understanding of equivalent expressions in radical expression in rational exponent form. This activity is designed to be done independently to gather accurate data on student mastery of the content of this lesson. Students should not use a calculator to complete the Wrap-Up.



1.6.7 Wrap-Up: Cool Down

1. Circle the expressions equivalent to $(25^3)^{\frac{1}{2}}$.

$(25)^{\frac{3}{2}}$
 $(\sqrt{25})^3$
 $(25)^{\frac{2}{3}}$
 $(25^{\frac{1}{3}})^2$
 $\sqrt{25^3}$

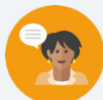


Student performance on this Wrap-Up can provide data that indicates additional support and differentiation that may be necessary. Use the responses to track any common errors made by the class. Consider using the most common errors in a “my favorite mistake” with the whole class or in small groups.

1.7 Evaluating Exponential Expressions

Lesson Introduction

 Benchmark	MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.		 Learning Targets	- I can evaluate numerical expressions involving rational exponents. - I can generate equivalent numerical expressions involving rational exponents.
 Benchmark Coherence	Building On MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.		Working Toward N/A	
 Essential Questions	- How can I use the Laws of Exponents to help me evaluate numerical expressions? - How can the relationship between rational exponent form and radical expression form help me evaluate numeric expressions without a calculator?			
 Lesson Narrative	In this lesson, students will extend their use of the Laws of Exponents to evaluate numerical expressions that are presented in rational exponent and/or radical expression form. Students will work together to practice how to evaluate these expressions and be able to do them on their own with accuracy by the end of the lesson.			
 Component Summary	1.7.1 Warm-Up (~5 min.)	Perfect squares puzzle activity (<i>SE p. 37</i>)	1.7.3 Activity (20 min.)	Equivalent expressions with exponents maze activity (<i>SE p. 38</i>)
	1.7.2 Collaboration (10 min.)	Exploring the simplification of expressions with exponents (<i>SE p. 37</i>)	1.7.4 Your Turn! (10 min.)	Evaluating expressions in rational exponent form without a calculator (<i>SE p. 39</i>)
	1.7.5 Wrap-Up (~5 min.)	Self-reflection activity about the things they are unsure about from the lesson and what they will do to improve (<i>SE p. 39</i>)		
 Resources	Glossary		Materials	
	Key Terms: N/A Additional Terms: - Rational Exponent - Radical Form - Equivalent Expression		N/A	
			Additional Preparation Work through the maze before you have students do it. When completing it, time yourself so you can determine an appropriate amount of time your students will need to complete it or get as far as they can.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not remember to convert between forms where appropriate. - Students may not make the connection between the two forms. - Students may put the numerator of the fraction in the exponent as the index of the radical form. - Students may make calculation errors when determining the value of the expression.	- This lesson requires students to do evaluation without a calculator and the use of perfect squares table. The table that will be created in question 1 of the 1.7.1 Warm-Up can be a valuable reference tool for visual learners in the future. - Consider having this table created for students to physically fill out and keep somewhere in their math tools folder or file for future use. As an alternative, consider creating the table on an Anchor Chart somewhere for students to reference when needed. - Consider discussing common errors from 1.6.6 Wrap-Up before 1.7.2 Collaboration. - Consider not allowing a calculator on the lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Which One Doesn't Belong? 1.7.3 Activity provides students with multiple scenarios where students must decide which one of the adjacent spaces contains an expression that is equivalent to the current space. Students can engage in this protocol to determine which expression does not belong, using the conditions of equivalency.	MA.K12.MTR.1.1: Participate in Effortful Learning 1.7.3 Activity provides students with a challenging maze where students must apply all they have learned about equivalent forms with exponents to determine the next step to take in the equivalency maze. To do this, students will need to modify methods and choose the best tools needed to persevere while solving this task.
 Differentiate Instruction	Consider placing students in groups and assigning groups to each number of the 1.7.2 Collaboration. By differentiating which groups engage with each portion of the activity, the lesson can become a jigsaw in which students are responsible for sharing their learning with others. For 1.7.1 Warm-Up, consider having the table from question 1 created for students to physically fill out and keep somewhere in their math tools folder or file for future use. 1.7.4 Your Turn! provides students an opportunity to show what they know related to the content of this lesson. How students perform on this activity can provide differentiation data for students who may need additional support.	
Lesson Notes:		

1.7.1 Warm-Up: Perfect Squares Puzzle

Component Overview: Students will complete a puzzle based on perfect squares.



1.7.1 Warm-Up: Perfect Squares Puzzle

- Complete the table by finding the square of each whole number.

Whole Number	Square	Whole Number	Square	Whole Number	Square
1	$1^2 = 1$	8	$8^2 = 64$	15	$15^2 = 225$
2	$2^2 = 4$	9	$9^2 = 81$	16	$16^2 = 256$
3	$3^2 = 9$	10	$10^2 = 100$	17	$17^2 = 289$
4	$4^2 = 16$	11	$11^2 = 121$	18	$18^2 = 324$
5	$5^2 = 25$	12	$12^2 = 144$	19	$19^2 = 361$
6	$6^2 = 36$	13	$13^2 = 169$	20	$20^2 = 400$
7	$7^2 = 49$	14	$14^2 = 196$	21	$21^2 = 441$

- Place a digit in each blue box so that the principal square root is a whole number, and the expression is as close to 0 as possible. Use only the digits 1 to 9, at most one time each.
Reminder: The principal square root is the positive square root.

$$\sqrt{\boxed{2} \boxed{5} \boxed{6}} - \sqrt{\boxed{8} \boxed{1}} - \sqrt{\boxed{9}}$$



The table that will be created in question 1 can be a valuable reference tool for visual learners in the future. Consider having this table created for students to physically fill out and keep somewhere in their math tools folder or file for future use. As an alternative, consider creating the table on an Anchor Chart somewhere for students to reference when needed.



Consider allowing students to work together on question 2 in this activity. Students may need a reminder of what the directions mean, as this is the second time, they have encountered this type of problem in the material so far.

1.7.2 Collaboration: Simplifying Expressions With Exponents

Component Overview: Students will work with a partner or small group to explore simplifying expressions with exponents. For this activity, release students to work with their partner for all five questions, prior to bringing the class back together to review the answers.

1.7.2 Collaboration: Simplifying Expressions with Exponents

1. With a partner, write the expression $(27)^{\frac{2}{3}}$ in three equivalent forms.

$(27)^{\frac{2}{3}}$	$(27^{\frac{1}{3}})^2$	$(\sqrt[3]{27})^2$	$\sqrt[3]{27^2}$
----------------------	------------------------	--------------------	------------------

2. Explain which equivalent form makes it easiest to find the value of $(27)^{\frac{2}{3}}$ without a calculator.

$(\sqrt[3]{27})^2$ allows me to determine the cube root of 27, which is 3 and then square that value which would be 9. Each of those calculations can be done using mental math.

3. Find the value of $(27)^{\frac{2}{3}}$ without a calculator.
9

4. Verify the value of $(27)^{\frac{2}{3}}$ using your calculator.
Students should be able to show the answer on their calculator is the same as their answer above.

5. For each expression, write an equivalent radical expression. Then, evaluate the expression using your calculator.

Expression	Equivalent Radical Expression	Value of the Expression
$(144)^{-\frac{3}{2}}$	$\frac{1}{\sqrt{144^3}}$	$\frac{1}{1,728}$
$-(-64)^{\frac{1}{3}}$	$-\sqrt[3]{-64}$	4
$(225)^{-\frac{1}{2}}$	$\frac{1}{\sqrt{225}}$	$\frac{1}{15}$
$-9^{\frac{5}{2}}$	$-\sqrt{9^5}$	-243
$(\frac{81}{256})^{\frac{3}{4}}$	$\sqrt[4]{(\frac{81}{256})^3}$	$\frac{27}{64}$



Students' fluency on equivalent forms should be increasing. As you circulate, look for students who still have to refer to previous lessons to know how to write equivalent expressions. Make note of these students and consider utilizing the activity to support these students using small group instruction. Consider asking these questions to help establish student understanding:



- How did you go about deciding which three equivalent forms you were going to use?
- Why did you choose the form you did for question 2?



Students may write different equivalent radical expressions. For some, determining the value from "their" equivalent expression may be difficult without a calculator. Instead of giving the student a calculator, ask the student to find another way to write their equivalent radical expression so that finding the value would be easier. For example, $(144)^{-\frac{3}{2}}$ written as $\frac{1}{\sqrt{144^3}}$ is harder than $\frac{1}{(\sqrt{144})^3}$. This helps the student understand why it is valuable to know how to rewrite numeric expressions with rational expressions using a variety of equivalent expressions.

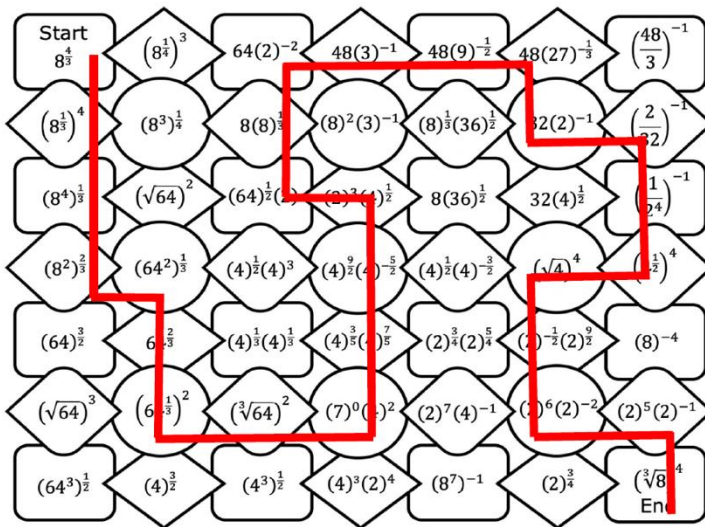
1.7.3 Activity: Exponents Maze

Component Overview: Students will complete a maze where they go from start to finish by only connecting to an adjacent equivalent exponent. For this activity, review the narrative information from the question prior to releasing students to do questions 1 and 2.



1.7.3 Activity: Exponents Maze

1. Work with a partner to complete the maze. Beginning at the "start" box, connect each expression to an equivalent expression to form one continuous path to the "end" of the maze. Move only horizontally or vertically.



2. What is the value of the expression in each box that created the path through the maze?

16



When reviewing the instructions and context for question 1, consider providing students with strategies to leverage when working together as partners. For example, each partner takes one of the adjacent expressions and evaluates them. Each partner shares what they evaluated to see if either of them gets the same answer as the prior expression. This will help students save time when trying to complete this activity.



For a challenge, allow students who are doing well with the content to work on this maze independently if time allows.



Consider using a timer during this activity. If left unstructured, this activity could take the remainder of the class period. Consider applying a time limit to this activity. The purpose is to provide practice of identifying equivalent expressions, not strictly completing the maze.

1.7.4 Your Turn!

Component Overview: Students will work independently to practice evaluating expressions in rational exponent form without a calculator. For this activity, release students to complete the problems in the table on their own. Bring students back together to review the answers to the activity.



1.7.4 Your Turn!

1. Evaluate each expression without a calculator.

$(100)^{\frac{1}{2}} \cdot (2^3)$ 80	$(8^{-1})\left(4^{\frac{5}{2}}\right)$ 4
$(81^6)^{\frac{1}{24}}$ 3	$(125)^{\frac{2}{3}}$ 25
$(49^{\frac{1}{2}})(3^{-2})$ $\frac{7}{9}$	$-(3^{12})^{\frac{1}{4}}$ -27
$(2^4)\left(8^{-\frac{2}{3}}\right)$ 4	$\left(\frac{15}{45}\right)^{-4}$ 81
$-\left(\frac{3}{48}\right)^{\frac{3}{2}}$ $-\frac{1}{64}$	$\left(\frac{144}{169}\right)^{-\frac{1}{2}}$ $1\frac{1}{12}$ or $\frac{13}{12}$



Consider using this time to work with a small group of students who may need additional support based on their performance throughout this lesson and prior lessons in this unit. In the small group, use squares 1 and 3 and allow students to work on squares 2 and 4 on their own and show you their answers and work so you can guide them through the process when needed.



Consider allowing students to check their work with a partner if they finish before the allotted time. During the share-out of the answers, have students explain how they got their answers and any methods or strategies that they leveraged.



- Which problem did you find the most challenging to evaluate?
- Were there any that, after seeing the correct answer, you are still unclear as to how that answer was correct?

1.7.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection activity about the things they are unsure about from the lesson and what they will do to improve. This activity should be done independently to get an accurate assessment of where students feel unsure related to the content of this lesson.



1.7.5 Wrap-Up: Reflection

Complete the following statements about today's lesson.

Answers will vary.

1. Today I am still unsure about

2. To fill in this gap I intend to



Since this Wrap-Up activity is about a student's self-reflection, this data can be coupled with their overall academic performance on this lesson to create lists of students, by combination profile, who may need additional differentiated support.

1.8 Equivalent Radical Expressions

Lesson Introduction

 Benchmark	MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.		 Learning Targets	- I can relate the power of a product law of exponents to the Product Law of Radicals - Given a numerical radical, I can use the product law of radicals to write an equivalent numerical radical.
 Benchmark Coherence	Building On MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.		Working Toward MA.912.NSO.1.5 Add, subtract, multiply, and divide algebraic expressions involving radicals.	
 Essential Questions	- How can the Power of a Product Law help me understand and use the Product Law of Radicals? - How does the Product Law of Radicals help me generate equivalent radical expressions?			
 Lesson Narrative	In this lesson, students will leverage their knowledge of the Laws of Exponents to learn more about the Product Law of Radicals. Students will learn to use the Product Law of Radicals to generate an equivalent radical expression when one is provided. They will have the opportunity to practice this skill throughout the lesson.			
 Component Summary	1.8.1 Warm-Up (~5 min.)	Exponents error analysis activity (<i>SE p. 40</i>)	1.8.4 Try It (5 min.)	Practice writing equivalent radical expressions (<i>SE p. 43</i>)
	1.8.2 Exploration (10 min.)	Exploration on generating equivalent rational expressions, given a rational expression (<i>SE p. 41</i>)	1.8.5 Your Turn! (5 min.)	Identifying and writing equivalent radical expressions (<i>SE p. 44</i>)
	1.8.3 Guided Instruction (20 min.)	Instruction on writing equivalent radical expressions (<i>SE p. 42</i>)	1.8.6 Wrap-Up (~5 min.)	Reflecting on most difficult part of the lesson (<i>SE p. 44</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Product Law of Radicals <u>Additional Terms:</u> - Rational Exponent - Radical Form - Equivalent Expression		N/A	
<u>Additional Preparation</u> N/A				

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not know their factor pairs well enough to rewrite radicals. - Students may miscalculate the equivalent radical expression. - Students may make mistakes when determining if a radical is equivalent.	- This lesson may cause barriers for students who lack strength in number sense. Consider potential interventions that may be necessary to help students who have significant learning deficits from prior grade levels. - Avoid the use of the word “simplify” when rewriting radical expressions. The purpose of rewriting radical expressions was because before calculators, books included a table of decimal equivalents to radical values. So, to determine an approximation of $\sqrt{150}$, the radical would have to be rewritten to $5\sqrt{6}$ and then the decimal equivalent of $\sqrt{6}$ found in the table, multiplying that by 5 to find the decimal equivalent. So $5\sqrt{6}$ is not “simpler” than $\sqrt{150}$. It is equivalent.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	True or False In 1.8.1 Warm-Up, students will see two explanations of the same exponential expression being evaluated. Consider leveraging the True or False protocol by having students determine whether the statements provided are true or false by using proper justification.	MA.K12.MTR.4.1: Discussion and Reflection In 1.8.1 Warm-Up, students will review the work of two students as they evaluate an exponential expression. Students will determine which student has made an error and describe how to correctly complete the task.
 Differentiate Instruction	Consider your partner pairings in this lesson. Due to the complex nature of this concept, consider grouping students together based on learning styles so they can connect over that commonality to work together towards understanding the concept. Where student work is provided for analysis or completion, consider highlighting each step in a different color and notating exactly what happened at each step. This will provide students with a visual reference for students to refer to when working on similar problems.	
Lesson Notes:		

1.8.1 Warm-Up: Error Analysis

Component Overview: Students will complete an error analysis activity on the work of two students' understanding of exponents. This activity is designed to be done independently to determine what students recall from prior lessons.



1.8.1 Warm-Up: Error Analysis

Leilana and Khamari wrote different numerical expressions and expanded them to show each has a value of 16.

Leilana's Work	Khamari's Work
$2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$	$2^8 = 2 \cdot 8 = 16$

1. Explain whether you agree with Leilana's work or Khamari's work.

Leilana is correct because the exponent of 4 indicates that the base of 2 should be multiplied by itself 4 times.

2. Write another numerical expression involving exponents with a value of 16.

4^2



Once students have completed the activity, allow students to vote to determine which student they believe made the error. Ask students to explain where the error was made, what the student may have been thinking, and how the problem should have been evaluated. For question 2, allow students to share out the numerical expressions that they came up with to get a value of 16.



- What misconception do you think Khamari had to come up with a value of 16?
- How would you prove to Khamari that the answer should be 256, and not 16?

1.8.2 Exploration: Equivalent Radical Expressions

Component Overview: Students will complete an exploration to discover how to generate equivalent rational expressions given a rational expression by analyzing two students work.



1.8.2 Exploration: Equivalent Radical Expressions

Ameer was asked to write an equivalent expression for $\sqrt{18}$. His work and resulting expression are shown.

Given Expression	Ameer's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (6 \cdot 3)^{\frac{1}{2}} = (6)^{\frac{1}{2}} \cdot (3)^{\frac{1}{2}}$	$\sqrt{6} \cdot \sqrt{3}$

Sophia wrote a different equivalent expression for $\sqrt{18}$. Her work and resulting expression are shown.

Given Expression	Sophia's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (9 \cdot 2)^{\frac{1}{2}} = 9^{\frac{1}{2}} \cdot 2^{\frac{1}{2}} = \sqrt{9} \cdot \sqrt{2}$	$3\sqrt{2}$

- What similarities and differences do you notice between Ameer's work and Sophia's work?

Similarities	Differences
Answers will vary. In both cases, they broke down the 18 into two factors.	Answers will vary. Both students simplified 18 into two different factors. Sophia used a factor that would result in a perfect square.



Be explicit with the language used when explaining Ameer and Sophia's work. Elicit responses from students to articulate what is occurring at each step so that they can clearly follow the structure and logic of each student's work in generating their equivalent expression.



During the student work review component, consider highlighting each step in a different color and notating exactly what happened at each step. This will provide students with a visual reference to refer to when working on similar problems.



- When you consider the similarities and the differences in each student's work, do their resulting expressions result in being equivalent to the given expression?
- How do you think that is possible?

1.8.2 Exploration: Equivalent Radical Expressions (continued)

2. With your partner, complete the work to find equivalent expressions for $\sqrt{175}$ in two different ways.

Given Expression	Work	Equivalent Radical Expression
$\sqrt{175}$	$175^{\frac{1}{2}} = (35 \cdot 5)^{\frac{1}{2}} = (35)^{\frac{1}{2}} \cdot (5)^{\frac{1}{2}}$	$\sqrt{35} \cdot \sqrt{5}$
$\sqrt{175}$	$175^{\frac{1}{2}} = (25 \cdot 7)^{\frac{1}{2}} = (25)^{\frac{1}{2}} \cdot (7)^{\frac{1}{2}}$	$\sqrt{25} \cdot \sqrt{7}$



Have students share out how they determined the missing pieces of the work that led to the equivalent radical expression.

3. Which equivalent radical expression above can be rewritten in the form of an **integer** multiplied by a **radical**?

$$\sqrt{25} \cdot \sqrt{7} = 5\sqrt{7} \text{ because } 25 \text{ is a perfect}$$

square, therefore $\sqrt{25} = 5$.

1.8.3 Guided Instruction: Equivalent Radical Expressions

Component Overview: Students will receive guided instruction on how to write equivalent expressions by using the power of a product and product law of radicals.



1.8.3 Guided Instruction: Equivalent Radical Expressions

Recall the **power of a product** law of exponents. It states the product of two or more numbers raised to a power is equal to the product of each number raised to the same power.

$$(ab)^m = a^m \cdot b^m \text{ or } a^m \cdot b^m = (ab)^m$$

The **product law of radicals** works similarly. Each of these equivalent expressions demonstrates the product law of radicals.

$$\begin{array}{ll} \sqrt{18} = \sqrt{6} \cdot \sqrt{3} & \sqrt{175} = \sqrt{35} \cdot \sqrt{5} \\ \sqrt{18} = \sqrt{9} \cdot \sqrt{2} & \sqrt{175} = \sqrt{25} \cdot \sqrt{7} \end{array}$$

When a radical expression is rewritten as a product of its factors, each of the radicals will have the same index.

Product Law of Radicals

$$\sqrt[m]{ab} = \sqrt[m]{a} \cdot \sqrt[m]{b} \text{ or } \sqrt[m]{a} \cdot \sqrt[m]{b} = \sqrt[m]{ab}$$

1. Rewrite $\sqrt[3]{325}$ in two different ways using the product law of radicals.

$$\sqrt[3]{65} \cdot \sqrt[3]{5}; \sqrt[3]{25} \cdot \sqrt[3]{13};$$

When rewriting radical expressions, mathematicians generally try to rewrite the expression in the form of an **integer** multiplied by a **radical**, when possible.

$$\sqrt{18} = \sqrt{9} \cdot \sqrt{2}, \text{ which can be rewritten as } 3\sqrt{2}$$

$$\sqrt{175} = \sqrt{25} \cdot \sqrt{7}, \text{ which can be rewritten as } 5\sqrt{7}$$

2. What was done to rewrite the example expressions?
A perfect square was used when choosing two factors.



When teaching the content prior to question 1, it is important to review the vocabulary. Consider leveraging any guided notes or vocabulary strategies when reviewing this material. Continue to use the language of radicals, factors, and index. Avoid short cuts of “twins”, “pairs”, or “buddies” when helping students’ understanding.



Have students explain how the Laws of Exponents, power of a product, and the product law of radicals are similar.

1.8.3 Guided Instruction: Equivalent Radical Expressions (continued)

3. The following radical expressions are equivalent to $\sqrt[3]{320}$.

$$\sqrt[3]{32} \cdot \sqrt[3]{10}$$

$$\sqrt[3]{64} \cdot \sqrt[3]{5}$$

$$\sqrt[3]{8} \cdot \sqrt[3]{40}$$

$$\sqrt[3]{16} \cdot \sqrt[3]{20}$$

- a. Determine which of the expressions can be rewritten in the form of an integer multiplied by a radical.

$$\sqrt[3]{64} \cdot \sqrt[3]{5} ; \sqrt[3]{8} \cdot \sqrt[3]{40}$$

- b. Discuss with your partner why these expressions can be rewritten.

Answers will vary. Each expression had a perfect cube, which would result in a constant multiplied by a radical.

4. Some radical expressions are given as an integer multiplied by the radical before any rewriting is done. Priya and Raj were asked to rewrite this type of expression. Their work is shown.

Priya's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 6\sqrt{5}$
Raj's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 5\sqrt{5}$

Explain whether Priya or Raj rewrote the expression correctly.

Priya is correct. The $3 \cdot 2$ portion of the process would result in 6, not 5.



After allowing students to try question 3, have students share out what answers they came up with for question 3a and chart those on the board to showcase all the additional equivalent expressions that were found by the class.



- *Where did Raj go wrong when simplifying the expression?*
- *How would you show/convince Raj that the work is incorrect?*



Allow students to discuss the answer in question 4 and share out how they know the work led to an incorrect final expression. Survey the class to determine who chose Priya as being correct and who chose Raj as being correct and use that data to generate a discussion on potential misconceptions in the class.

1.8.4 Try It: Equivalent Radical Expressions

Component Overview: Students will practice writing equivalent radical expressions. Students are not expected to have full mastery yet. The teacher should be looking for misconceptions during this time.



1.8.4 Try It: Equivalent Radical Expressions

- Write two equivalent expressions for each radical expression. Show all your steps.

a.

$\sqrt{135}$
$\sqrt{135} = \sqrt{27} \cdot \sqrt{5}$
$\sqrt{135} = \sqrt{9} \cdot \sqrt{15} = 3 \cdot \sqrt{15} = 3\sqrt{15}$

b.

$\sqrt[3]{600}$
$\sqrt[3]{600} = \sqrt[3]{100} \cdot \sqrt[3]{6}$
$\sqrt[3]{600} = \sqrt[3]{300} \cdot \sqrt[3]{2}$

- Write an equivalent expression for the following radical expression.

$$5\sqrt[3]{56}$$

$$10\sqrt[3]{7}$$



Allow students to try both questions on their own, providing support to students as you facilitate this activity. If students finish prior to the time limit you have set, consider allowing them to check their work with a partner prior to moving to a whole-class review of the answer.



For a challenge, consider having students generate an additional equivalent expression for the provided radical expression.

1.8.5 Your Turn!

Component Overview: Students will work independently to practice identifying and writing equivalent radical expressions. For this activity, release students to work on question 1 and 2 on their own, and then call the class back together to review the correct answers.



1.8.5 Your Turn!

1. Select all expressions that are equivalent to $\sqrt{40}$.

- ☐ $(40)^{\frac{1}{2}}$
- ☐ $2\sqrt{20}$
- ☐ $\sqrt{4} \cdot \sqrt{10}$
- ☐ $(20 \cdot 2)^{\frac{1}{2}}$
- ☐ $5\sqrt{8}$
- ☐ $2\sqrt{10}$
- ☐ $(10 \cdot 4)^{\frac{1}{2}}$

2. Write an equivalent expression for $\sqrt[3]{378}$ in the form of a constant multiplied by a radical.

$$3\sqrt[3]{14}$$



Remind students of questions that use the phrase “select all.” Remind them to review each answer choice intently to determine if it meets the criteria for equivalency and would be an answer that could be selected. Students may forget that there can be more than one correct answer to these question types.



For a challenge, consider having students create an additional expression that would be a correct answer choice for question 1.

1.8.6 Wrap-Up: Reflection

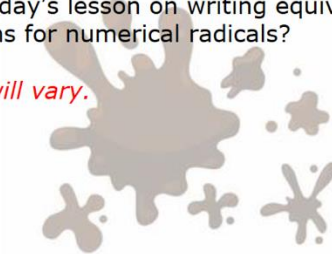
Component Overview: Students will complete a Ticket Out the Door that assesses students on their understanding of identifying and writing equivalent radical expressions. This activity should be done independently to provide the most accurate student-level data.



1.8.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on writing equivalent expressions for numerical radicals?

Answers will vary.



Consider allowing students to choose how they respond to the reflection. For instance, students can create a video, audio file, graphic organizer, etc. Encourage students to reference specific examples from the lesson in their responses.

1.9 Adding and Subtracting Radical Expressions

Lesson Introduction

 Benchmark	MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.			 Learning Target	I can add or subtract two square roots or two cube roots.
 Benchmark Coherence	Building On			Working Toward	
	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.			MA.912.NSO.1.5 Add, subtract, multiply, and divide algebraic expressions involving radicals.	
 Essential Questions	- How do I add or subtract two radical expressions? - What is the process for determining if two radical expressions can be added?				
 Lesson Narrative	In this lesson, students will extend their knowledge of radical expressions to add and subtract them. Students will experience situations where the radical expression contains the same radical values and situations where they contain different radical values.				
 Component Summary	1.9.1 Warm-Up (~5 min.)	Adding and Subtracting Algebraic Expressions Activity (<i>SE p. 45</i>)	1.9.4 Guided Practice (5 min.)	Adding and subtracting radical expressions practice activity (<i>SE p. 48</i>)	
	1.9.2 Exploration (5 min.)	Exploring the addition and subtraction of radical expressions (<i>SE p. 46</i>)	1.9.5 Your Turn! (5 min.)	Independent practice finding the sum and/or difference of radical expressions (<i>SE p. 49</i>)	
	1.9.3 Guided Instruction (20 min.)	Adding and subtracting radical expressions instructional activity (<i>SE p. 46</i>)	1.9.6 Wrap-Up (~5 min.)	Ticket Out the Door on finding the sum of a radical expression (<i>SE p. 49</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Like Radicals <u>Additional Terms:</u> - Rational Exponent - Radical Form - Equivalent Expression		N/A		Consider potential interventions that may be necessary based on a student's unfinished learning from a prior lesson or prior grade level. For example, see the differentiation note for the 1.9.1 Warm-Up activity. If students struggle with question 1, they may need intervention related to combining like terms.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not realize that a radical can be simplified in order to be added. - Students may make calculation errors when adding or subtracting. - Students may not know their factor pairs well enough to simplify the radicals.	This lesson incorporates several “select all that apply” style questions. Consider reminding students that these types of questions can appear on high-stakes tests and encourage students to critically think through each answer choice prior to completing the entire question. “Select all” questions are often a good way to assess fluency and/or equivalency.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	True or False In 1.9.4 Guided Practice, students will choose all the expressions that could be combined. Consider leveraging a True or False routine for this activity. Present each answer choice and state that they can be combined. Have students determine if that statement is true or if it is false, discussing why it is either answer.	MA.K12.MTR.4.1: Discussion and Reflection In 1.9.2 Exploration, students will review the work of two students as they add two rational expressions. Students will determine which student has made an error and describe how to correctly complete the task.
 Differentiate Instruction	Consider partnering students up based on their performance in the prior lesson, as students will have different entry points to this lesson. Consider having a high-medium and medium-low pairing scenario where students who performed well with the previous lesson are paired with those who did not. The knowledge of the prior lesson is essential to success in this one. 1.9.3 Guided Instruction provides students with three steps to be used when adding and subtracting rational expressions. Consider creating a visual aid such as an anchor chart to post these three steps for students to reference. On the visual aid, show at least three examples of radical expressions being added and subtracted following each of these three steps. Students will complete a Ticket Out the Door that assesses student understanding of adding and subtracting radical expressions. Student performance on this activity can provide data for potential differentiation that may be needed for specific students.	
Lesson Notes:		

1.9.1 Warm-Up: Adding and Subtracting Algebraic Expressions

Component Overview: Students will complete a series of questions that have them combine like terms provided in different forms, then students make a prediction about like terms of radical expressions that will be explored during the lesson.



1.9.1 Warm-Up: Adding and Subtracting Algebraic Expressions

1. Find the sum or difference of each algebraic expression.

$$\begin{array}{r} 4x + 23x \\ 27x \end{array}$$

$$\begin{array}{r} -7t - 40t \\ -47t \end{array}$$

$$\begin{array}{r} -35b + 10b \\ -25b \end{array}$$

2. Complete the statement to explain why it is possible to combine the terms in each of these expressions.

When adding and subtracting algebraic expressions, **like** terms can be combined.

3. Which, if any, of the radical expressions do you think can be combined?

- ☐ $\sqrt{17} + \sqrt{32}$
- ☐ $5\sqrt{13} - 8\sqrt{3}$
- ☒ $-7\sqrt{17} + 3\sqrt{17}$
- ☒ $9\sqrt{5} + \sqrt{5}$
- ☐ $8\sqrt{40} - 4\sqrt{9}$
- ☒ $11\sqrt{48} - 6\sqrt{48}$
- ☐ $-2\sqrt{49} - 3\sqrt{4}$



Watch for student performance on questions 1 and 2. If students are struggling to get the correct answers for question 1, there may be need for additional prerequisite intervention. Understanding the concept of like terms and being able to combine them appropriately will be a key skill for adding and subtracting radical expressions.



- How did you determine which radical expressions in question 3 could be combined?
- What is the relationship between what you know about like terms and the expressions in question 3?

1.9.2 Exploration: Adding and Subtracting Radical Expressions

Component Overview: Students will work in small groups to explore the addition and subtraction of radical expressions. For this activity, release students to complete questions 1 – 4 with their small groups. Bring the class back together and go through each question as an entire class.



1.9.2 Exploration: Adding and Subtracting Radical Expressions

1. In small groups, determine the value of each expression.

$$\sqrt{25} + \sqrt{100}$$

15

$$\sqrt{25 + 100}$$

11.1803

2. Explain which of the expressions is equivalent to $\sqrt{125}$.
The value of $\sqrt{25 + 100}$ is equivalent to $\sqrt{125}$ because the values under the radicand must be combined first.

3. Paloma says $\sqrt{16} - \sqrt{9} = \sqrt{7}$. Ty'rique says $\sqrt{16} - \sqrt{9} = 1$. Who is correct? Verify by entering $\sqrt{16} - \sqrt{9}$ into a calculator.

Ty'rique is correct because $\sqrt{16} - \sqrt{9}$ is equal to 1.

4. Xavier says $\sqrt{12} + 2\sqrt{12} = 3\sqrt{12}$. Sam says $\sqrt{12} + 2\sqrt{12} = 2\sqrt{24}$. Who is correct? Verify using a calculator.

Xavier is correct because $\sqrt{12} + 2\sqrt{12}$ is equal to $3\sqrt{12}$.



Students will need to do verification of values of radical expressions. Some students may need a refresher of how to use their calculator to do this appropriately. Consider using an on-screen calculator or document camera to show the steps to evaluating a radical expression.



Once the activity is completed, allow students to share out their responses and methods for each question. Consider polling the class to determine how many students arrived at the same conclusion. Have students vote on which students they believe were correct in questions 3 and 4, sharing their reasoning for their answer.



When verifying using a calculator, what do you notice when taking the square root of 7 vs taking the square root of 16 and subtracting the square root of 9?

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions

Component Overview: Students will receive guided instruction on how to add and subtract radical expressions using like radicals and rewriting radical expressions in order to add or subtract like radicals.



1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions

When adding and subtracting algebraic expressions, only **like terms** can be combined. Adding and subtracting radical expressions is similar because only like radicals can be combined.



Like radicals are radical terms that have the same index and the same radicand.

1. Find the sum or difference of each radical expression.

a. $4\sqrt{5} + 7\sqrt{5} = 11\sqrt{5}$

b. $30\sqrt[3]{2} - 23\sqrt[3]{2} = 7\sqrt[3]{2}$

c. $-10\sqrt{7} - 23\sqrt{7} = -10\sqrt{7} - 23\sqrt{7}$

d. $-\sqrt{17} + \sqrt{17} = 0$

Three steps can be used when adding and subtracting radical expressions.

Step 1: Find perfect squares (for square roots) or perfect cubes (for cube roots) that are factors of the radicand.

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares or cubes.

Step 3: Add or subtract like radical terms.



- What is the relationship between each of these radicals in question 1?
- Can you work with like radicals the same way you can with like terms? Why or why not?



Consider creating a visual aid such as an anchor chart to post these three steps for students to reference. On the visual aid, show at least three examples of radical expressions being added and subtracted following each of these three steps.

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions (continued)

2. Use the steps to add the following radical expression.

$$3\sqrt{6} + 5\sqrt{24}$$

Step 1: Both terms include square roots. If possible, find perfect squares that are factors of the radicands, 6 and 24.

$$6 = 3 \cdot 2$$

$$24 = 6 \cdot 4$$

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares.

$$3\sqrt{6} + 5\sqrt{24} = 3\sqrt{6} + 10\sqrt{6}$$

Step 3: Add or subtract like radical terms.

$$13\sqrt{6}$$

3. Three students added $\sqrt{180}$ and $\sqrt{45}$ differently. Their work is shown.

Gerald's Work	Kalo's Work	Paola's Work
$\sqrt{180} + \sqrt{45}$	$\sqrt{180} + \sqrt{45}$	$\sqrt{180} + \sqrt{45}$
$2\sqrt{45} + \sqrt{45}$	$3\sqrt{20} + 3\sqrt{5}$	$6\sqrt{5} + 3\sqrt{5}$
$3\sqrt{45}$		$9\sqrt{5}$

- a. Use a calculator to verify if the students' answers are equivalent.

All three are equivalent.

- b. Why did the students end up with different expressions for their sum?

The students chose different factors.



When guiding students through question 2, be sure to teach each step explicitly, showing students how that step is being applied to this specific question. Have students discuss why each step is necessary in this problem and how to verify the step is done correctly.



Explain if steps 1 and 2 are necessary for every situation where you must add or subtract rational expressions.



Consider having students share their methods for question 3a. Use an on-screen calculator or a document camera to have students use their calculator to walk the class through the steps they used to verify each student's work.

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions (continued)

- c. Show how Gerald and Kalo could continue their work to arrive at the same expression as Paola.

Gerald's Next Steps	Kalo's Next Steps
$3\sqrt{45}$	$3\sqrt{20} + 3\sqrt{5}$
$3 \cdot \sqrt{9} \cdot \sqrt{5}$	$3 \cdot \sqrt{4} \cdot \sqrt{5} + 3\sqrt{5}$
$3 \cdot 3 \cdot \sqrt{5}$	$3 \cdot 2 \cdot \sqrt{5} + 3\sqrt{5}$
$9 \cdot \sqrt{5}$	$6 \cdot \sqrt{5} + 3\sqrt{5}$
$9\sqrt{5}$	$6\sqrt{5} + 3\sqrt{5}$
	$9\sqrt{5}$

4. Megh showed her work on the problem.

$$\sqrt[3]{32} + \sqrt[3]{64} = \sqrt[3]{8 \cdot 4} + 4 = 2\sqrt[3]{4} + 4 = 6\sqrt[3]{4}$$

Write an explanation to Megh explaining the error in her work.

The final answer should have been left at $2\sqrt[3]{4} + 4$. These terms are not like terms and can not be combined.



Have students share out their answer to question 3c. Consider having a student come to the board for each student and showing how to complete the remainder of the steps to arrive at the same answers as Paola. Have students not at the board offer feedback and support for the student at the board.

As students gain in their proficiency, release the students from showing work. Students do not have to show all the steps that are shown in the key.



- What makes the expression in question 4 different from the others that you have experienced?
- What about the process, if anything, was different from the other expressions?

1.9.4 Guided Practice: Adding and Subtracting Radical Expressions

Component Overview: Students will practice adding and subtracting radical expressions in a structured format. For this activity, release students to work with a partner to answer each question.



1.9.4 Guided Practice: Adding and Subtracting Radical Expressions

1. Select all of the expressions that can be combined. For the selected expressions, combine the terms. Show your work.

☐ $\sqrt{17} + \sqrt{32}$

☐ $5\sqrt{13} - 8\sqrt{3}$

☒ $-7\sqrt{17} + 3\sqrt{17}$

☒ $9\sqrt{5} + \sqrt{5}$

☐ $8\sqrt{40} - 4\sqrt{9}$

☒ $11\sqrt{48} - 6\sqrt{48}$

☒ $-2\sqrt{49} - 3\sqrt{4}$

With a partner, find the sum and difference of the following problems. If you cannot add or subtract, circle the expression.

a. $\sqrt[3]{6} + 3\sqrt{6}$

c. $16^{\frac{1}{3}} - 2^{\frac{1}{3}}$
 $2^{\frac{3}{2}} - \sqrt[3]{2} = \sqrt[3]{2}$

b. $\sqrt{20} - 3\sqrt{45}$
 $-7\sqrt{5}$

d. $\sqrt{72} + \sqrt{18} - \sqrt{15}$
 $9\sqrt{2} - \sqrt{15}$



Remind students that this is an example of a “select all that apply” question. Students should go through each answer choice to determine if it can be combined using what they’ve learned in this lesson.



When reviewing the answers, consider leveraging a True or False protocol. Present each answer choice and state that they can be combined. Have students determine if that statement is true or if it is false, discussing why it is either answer.



For a challenge, have students create a three-term radical expression that, when simplified, will reduce to a single-term radical expression. Have them show their work to prove that this will happen. Also, consider having students trade their expression with a partner, and have them evaluate to determine if it reduces to a single term or not.

1.9.5 Your Turn!

Component Overview: Students will work independently on practicing find the sum and/or difference of radical expressions. For this activity, release students to try these questions independently and bring students back together to review answers.



1.9.5 Your Turn!

1. Find the sum or difference. If you cannot add or subtract, circle the expression.

a. $\sqrt[3]{27000} - \sqrt[3]{512}$
22

b. $\sqrt{50} + \sqrt{18} + \sqrt{10}$
 $8\sqrt{2} + \sqrt{10}$

c. $\sqrt[3]{6} - \sqrt[3]{48}$
 $-\sqrt[3]{6}$

d. $32^{\frac{1}{2}} - 16^{\frac{1}{3}}$



Based on student performance in the prior activities, consider creating a small group for students who need differentiated support. With the small group, guide students through question 1a and have them do question 1c and 1d. If time permits, walk students through question part b. The goal of this group is to reteach the content for mastery, not necessarily to complete all problems.



Consider calling students to the board to show their work for each question. Allow students to provide feedback to their peers and agree or disagree with the final answers and process used to get there.



For a challenge, allow students to create an addition or subtraction problem and trade it with a partner, if time permits. Have the partner find the sum or difference and compare their answer to the author of the original problem and discuss the answer.

1.9.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses student understanding of finding the sum of a radical expression. This activity should be done independently to gather the most accurate data. Students should not use a calculator to complete the Wrap-Up.



1.9.6 Wrap-Up: Ticket Out the Door

- Find the sum.

$$17\sqrt{8} + 5\sqrt{8}$$

$$44\sqrt{2}$$



Student performance on this Wrap-Up can provide data that indicates additional support and differentiation that may be necessary.

Pay attention to what students put as their final answer. For example, if students put $22\sqrt{40}$, the student misunderstands the “like terms” component of this type of problem.

1.10 Multiplying and Dividing Radical Expressions

Lesson Introduction

 Benchmark	MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.		 Learning Target	I can multiply and divide two square roots or two cube roots.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.		MA.912.NSO.1.5 Add, subtract, multiply and divide algebraic expressions involving radicals.	
 Essential Questions	- How do I multiply or divide two radical expressions? - When is it appropriate to rationalize the denominator?			
 Lesson Narrative	In this lesson, students will extend their knowledge of radical expressions to multiply and divide them. Students will get the opportunity to learn and practice this concept with radical expressions presented in multiple forms. Additionally, students will be introduced to the formal definition and process of rationalizing the denominator.			
 Component Summary	1.10.1 Warm-Up (~5 min)	Radical expression equivalency number puzzle (<i>SE p. 50</i>)	1.10.3 Activity (10 min)	Multiplying and dividing radical expressions card activity (<i>SE p. 53</i>)
	1.10.2 Guided Instruction (25 min)	Dividing and multiplying radical expressions (<i>SE p. 51</i>)	1.10.4 Your Turn! (5 min)	Multiplying and dividing radical expressions practice activity (<i>SE p. 53</i>)
	1.10.5 Wrap-Up (~5 min)	Level of confidence self-assessment related to multiplying and dividing two square roots or two cube roots (<i>SE p. 54</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Denominator - Rationalizing the Denominator <u>Additional Terms:</u> - Radical Form - Equivalent Expression		(Optional) Index cards or paper to make passable cards for the 1.10.3 Activity: Pass Around	Work through each of these problems prior to teaching. Highlight potential misconceptions that students may have to front-load some of your instructional support provided during the lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to rewrite the radical. - Students may not know their factor pairs well enough to simplify the radicals.	1.10.3 Activity requires students to complete a card passing activity. Consider your student learning styles to determine if you would like to prepare the cards ahead of time, use Anchor Charts, or sheets of paper. - Students should not be required to rationalize the denominator. The skill of rationalizing the denominator of an expression with a radical in the denominator was only necessary before students had access to calculators. Knowing that rationalizing the denominator is possible and results in an equivalent expression should be the reason students are asked to rationalize the denominator when a radical is in the denominator of a fraction.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Poll the Class 1.10.2 Guided Instruction allows students to choose a method and explain why they chose it. Consider leveraging the Poll the Class protocol to poll the class and determine who has chosen the same method, and if any of the students who chose the same method may have different rationale.	MA.K12.MTR.1.1: Participate in Effortful Learning 1.10.3 Activity has students working in groups to complete a step when multiplying or dividing radical expressions. Each student will complete a step and pass it to the next student for review and movement to the next step.
	For 1.10.2 Guided Instruction (question 4), allow for additional physical movement for kinesthetic learners. Consider designating one side of the room as “agree” and one side as “disagree”. Have students choose a side of the room in relation to Jean’s explanation and have a representative from each group explain why they agree or disagree with Jean’s explanation. Repeat this process for Marcus’s explanation as well.	
	For 1.10.3 Activity, consider having physical cards made for this activity for a hands-on experience for kinesthetic learners. These cards can be made by printing the page and cutting them out or putting the problems directly onto index cards. Another alternative is to put the problems on chart paper and have students rotate to complete the next step of the given problem on the chart.	
Review how students are doing in 1.10.4 Your Turn! to determine if additional support may be necessary after this lesson has been completed. This data can also be compiled with their self-reflection data provided in 1.10.5 Wrap-Up.		
Lesson Notes:		

1.10.1 Warm-Up: True Equations

Component Overview: Students will complete a number puzzle that leverages their knowledge of equivalent radical expressions.



1.10.1 Warm-Up: True Equations

- Using the digits 0 to 9, place a digit in each blue box to make both equations true. Each digit can only be used one time, at most.

$$\sqrt{\boxed{1}\boxed{8}} = \boxed{3} \sqrt{\boxed{2}}$$

$$\sqrt{\boxed{4}\boxed{9}} = \boxed{7}$$



Consider reminding students how to approach this type of problem with the use of digits 0 to 9 only once. Remind them of the work they have done with this in prior lessons and to leverage the tools and resources they have at their disposal to help them solve this problem.



Upon completion of the activity, allow students to share their answers to provide them with the correct answer. Survey the students to determine the different methods, strategies, and approaches used to attack this problem. Have students share if they used any of the same strategies. Invest the time in sharing strategies and ways of thinking so that it can support student thinking as they progress through the course.

1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions

Component Overview: Students will receive guided instruction on how to divide and multiply radical expressions by rewriting radicals and using the Laws of Exponents.



1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions

Recall the **quotient of powers** law of exponents.

$$\frac{a^m}{a^n} = a^{m-n}$$

Ariana used the **quotient of powers** law of exponents to simplify $\frac{\sqrt{10}}{\sqrt{10}}$. Her work is shown.

$$\frac{\sqrt{10}}{\sqrt{10}} = \frac{10^{\frac{1}{2}}}{10^{\frac{1}{2}}} = 10^{\frac{1}{2} - \frac{1}{2}} = 10^0 = 1$$

When dividing radicals with the same index, the **quotient law of radicals** is used to rewrite the expression. Each of the following equivalent expressions demonstrates the quotient law of radicals.

$$\frac{\sqrt{13}}{\sqrt{13}} = \sqrt{\frac{13}{13}} = \sqrt{1} = 1 \quad \frac{\sqrt{21}}{\sqrt{21}} = \sqrt{\frac{21}{21}} = \sqrt{1} = 1$$

1. Rewrite $\frac{\sqrt[3]{2}}{\sqrt[3]{2}}$ using the quotient law of radicals.

$$\sqrt[3]{\frac{2}{2}} = \sqrt[3]{1} = 1$$

When dividing radical expressions, rewrite radical expressions in equivalent forms and use the quotient law to eliminate radicals where possible.



How are the quotient of powers and the quotient law of radicals similar?



At this point in the lesson, don't introduce the concept of "rationalizing the denominator". This activity is designed to conceptually connect this concept for students to what they have previously learned. At the end of this lesson, students will be introduced to the formal concept of rationalizing the denominator.

1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions (continued)

2. Write an equivalent expression for $\frac{\sqrt{125}}{\sqrt{245}}$.

$\frac{\sqrt{125}}{\sqrt{245}}$	
$\frac{\sqrt{125}}{\sqrt{245}} = \frac{\sqrt{25 \cdot 5}}{\sqrt{49 \cdot 5}}$	The radicands 125 and 245 are rewritten as the product of a factor and 5.
$\frac{\sqrt{25 \cdot 5}}{\sqrt{49 \cdot 5}} = \frac{5 \cdot \sqrt{5}}{7 \cdot \sqrt{5}}$	The radicals are rewritten using the product law of radicals.
$\frac{5 \cdot \sqrt{5}}{7 \cdot \sqrt{5}} = \frac{5}{7} \cdot \frac{\sqrt{5}}{\sqrt{5}}$	Rewrite each perfect square root.
Because $\frac{\sqrt{5}}{\sqrt{5}} = 1$, $\frac{5}{7} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{5}{7}$.	



When providing instruction during question 2, allow students to respond to the fill-in-the-blank sections within the table. Ask students to explain where they got the value from and how they know it is correct. Once you have reviewed the table, ask students to summarize the process that occurred during the table, using a different example if necessary.

3. With your partner, rewrite the expression $\frac{\sqrt{275}}{\sqrt{44}}$ using the quotient laws of radicals.

$$\frac{\sqrt{275}}{\sqrt{44}} = \frac{5\sqrt{11}}{2\sqrt{11}} = \frac{5}{2}$$

4. Jean and Marcus were asked to explain how they would multiply $\sqrt{8} \cdot \sqrt{8}$. Their explanations are shown.

Jean's Explanation	I would first rewrite the radical expressions using rational exponents and then use the product of powers law of exponents to multiply the like bases. $\sqrt{8} \cdot \sqrt{8} = 8^{\frac{1}{2}} \cdot 8^{\frac{1}{2}} = 8^{\frac{1}{2} + \frac{1}{2}} = 8^{\frac{2}{2}} = 8^1 = 8$
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1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions (continued)

Marcus's Explanation

I would use the **product law of radicals** to multiply the radical expressions and then use the law again to rewrite the square root.

$$\sqrt{8} \cdot \sqrt{8} = \sqrt{8 \cdot 8} = \sqrt{64} = 8$$

Do you agree that both expressions have a value of 8?

Answers will vary. Yes, I agree because I determined the values were the same in my calculator. Yes, because in both methods, the final value is the same.

5. Explain which method you would use to multiply $\sqrt{12} \cdot \sqrt{12}$.

Answers will vary. I would use the product law of radicals because $\sqrt{12} \cdot \sqrt{12} = \sqrt{144} = 12$.

6. Multiply the following radical expressions.

$$\sqrt{6} \cdot \sqrt{18}$$

$6\sqrt{3}$



In cases where there is a fraction with a radical in the **denominator**, mathematicians use a technique called **rationalizing the denominator** to rewrite an expression.

A model of how to rationalize the expression $\frac{30}{\sqrt{2}}$ is shown.

$$\frac{30}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{30\sqrt{2}}{\sqrt{4}} = \frac{30\sqrt{2}}{2} = 15\sqrt{2}$$



To allow for additional physical movement for kinesthetic learners, consider a physical representation of student response. For example, designate one side of the room as “agree” and one side as “disagree.” Have students choose a side of the room in relation to Jean’s explanation and have a representative from each group explain why they agree or disagree with Jean’s explanation. Repeat this process for Marcus’s explanation as well.



For a challenge, consider the following additional question:

- Another student comes along and says they think both Jean and Marcus are wrong. How can you use your calculator to prove that both Jean and Marcus are correct in their final evaluation?

Have students explain, using their calculator, how they know both are correct.



Students can leave their answer to question 6 as $\sqrt{108}$. Students may also provide other equivalent answers.



Spend a moment connecting what students can do in this lesson with the formal vocabulary of “rationalizing the denominator.” If using a word wall or vocabulary space in the classroom, consider adding this phrase and definition to that space.

1.10.3 Activity: Pass Around

Component Overview: Students will complete a card passing activity where students will begin a multiplication or division of radical expressions, and other students complete their step. Students may also start the problem on a sheet of regular notebook paper and pass along for the same effect. Students will need to be put into groups of four in this activity.



1.10.3 Activity: Pass Around

In groups of four, each student completes the **first** step of their problem. The students then pass their problem to the student on their right to complete the next step. For each new step, continue passing the problems to the student on the right.

Student A $\frac{\sqrt[3]{200}}{\sqrt[3]{625}}$ $\frac{2\sqrt[3]{5}}{5}$	Student B $\sqrt[3]{14} \cdot \sqrt[3]{20}$ $2\sqrt[3]{35}$
Student C $\sqrt{24} \cdot \sqrt{30}$ $12\sqrt{5}$	Student D $\frac{\sqrt{256}}{\sqrt{108}}$ $\frac{8\sqrt{3}}{9}$



Consider having physical cards made for this activity for a hands-on experience for kinesthetic learners. These cards can be made by printing the page and cutting them out or putting the problems directly onto index cards. Another alternative is to put the problems on chart paper and have students rotate to complete the next step of the given problem on the chart.



Have students explain their steps to the other group members after writing the next step. Other members of the group should provide feedback and agree or disagree with the student's next step.



For a challenge, consider adding a blank question where each student creates a problem, and the process repeats itself for the newly created question.

1.10.4 Your Turn!

Component Overview: Students will work independently to practice multiplying and dividing radical expressions. For this activity, students should be released to work independently.



1.10.4 Your Turn!

1. Write an equivalent expression for each radical expression.

a. $\sqrt{9} \cdot \sqrt{9}$

9

b. $\frac{\sqrt[3]{56}}{\sqrt[3]{77}}$

$\frac{2\sqrt[3]{121}}{11}$

c. $\frac{2\sqrt{22}}{\sqrt{8}}$

$\sqrt{11}$

2. Anjelica was asked to rewrite the expression $\frac{\sqrt[3]{264}}{\sqrt[3]{81}}$.

Anjelica's work is shown. Write an explanation to Anjelica describing why her problem is incorrect. Then, show how to correctly work the problem.

$$\frac{\sqrt[3]{264}}{\sqrt[3]{81}} = \frac{\sqrt[3]{8 \cdot 33}}{\sqrt[3]{9 \cdot 9}} = \frac{2\sqrt[3]{33}}{9}$$

Anjelica, it looks like you treated the denominator as though it was a square root when it is a cube root.

$$\frac{\sqrt[3]{264}}{\sqrt[3]{81}} = \frac{\sqrt[3]{8 \cdot 33}}{\sqrt[3]{27 \cdot 3}} = \frac{2\sqrt[3]{33}}{3\sqrt[3]{3}} = \frac{2\sqrt[3]{11}}{3}$$



For students who were struggling in the prior activities in this unit, consider pulling into a small group for differentiated support based on what has been observed. Spend more time on question 1 with the small group to allow them additional instruction and practice on multiplying and dividing radical expressions.



Have students share out their methods for question 2. Consider having students come to the board to show their corrected steps. Ask students to articulate the misconception that Anjelica had and how they would approach teaching her how to correct it.



For a challenge, consider having students create an additional problem similar to question 1 and trade with a partner to solve.

1.10.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-assessment on their level of confidence related to multiplying and dividing two square roots or two cube roots. This activity is designed to be done independently to review confidence levels for each individual student.

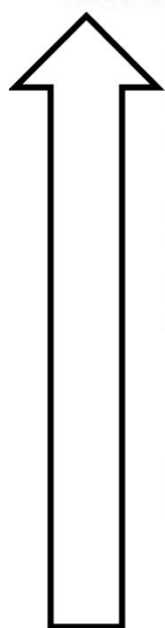


1.10.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of the following.

I can multiply and divide two square roots or two cube roots.

Answers will vary.



I am so confident; I could explain this to someone else!

I can get to the right answer, but I do not understand well enough to explain it yet.

I understand some of this, but I do not understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will get help by

I do not understand any of this yet. I will get help by



Student response to these questions can provide an insight into their level of confidence on multiplying and dividing two square roots or two cube roots. This data can be coupled with student performance to help create potential responses for how to better support each individual student.